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(12) **United States Patent**
Bei(10) **Patent No.: US 12,398,181 B2**(45) **Date of Patent: Aug. 26, 2025**(54) **METHODS AND COMPOSITIONS FOR
DELIVERY OF AGENTS ACROSS THE
BLOOD-BRAIN BARRIER**(71) Applicant: **The Brigham and Women's Hospital,
Inc.**, Boston, MA (US)(72) Inventor: **Fengfeng Bei**, Boston, MA (US)(73) Assignee: **The Brigham and Women's Hospital,
Inc.**, Boston, MA (US)(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 974 days.(21) Appl. No.: **17/258,846**(22) PCT Filed: **Jul. 11, 2019**(86) PCT No.: **PCT/US2019/041386**

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(2) Date: **Jan. 8, 2021**(87) PCT Pub. No.: **WO2020/014471**PCT Pub. Date: **Jan. 16, 2020**(65) **Prior Publication Data**

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(2013.01); **A61K 35/761** (2013.01); **A61K**
38/45 (2013.01); **A61K 48/0008** (2013.01);
A61K 49/0045 (2013.01); **A61K 49/0097**
(2013.01); **A61P 35/00** (2018.01); **C07K**
14/001 (2013.01); **A61K 38/00** (2013.01);
C07K 2319/00 (2013.01); **C12N 2750/14122**
(2013.01); **C12N 2750/14143** (2013.01)(58) **Field of Classification Search**CPC **C07K 14/005**; **C07K 14/001**; **C07K**
2319/00; **A61K 31/522**; **A61K 35/761**;
A61K 38/45; **A61K 48/0008**; **A61K**
49/0045; **A61K 49/0097**; **A61K 38/00**;
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See application file for complete search history.

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ABSTRACTThe present invention is based on the development of
artificial targeting sequences that enhance permeation of
agents into cells and across the blood brain barrier, compo-
sitions comprising the sequences, and methods of use
thereof. Provided herein is an AAV comprising a capsid
protein comprising a targeting sequence and a transgene,
preferably a therapeutic or diagnostic transgene. Further,
provided herein are methods of delivering a transgene to a
cell, the method comprising contacting the cell with an AAV
or fusion protein described herein.**11 Claims, 33 Drawing Sheets****Specification includes a Sequence Listing.**

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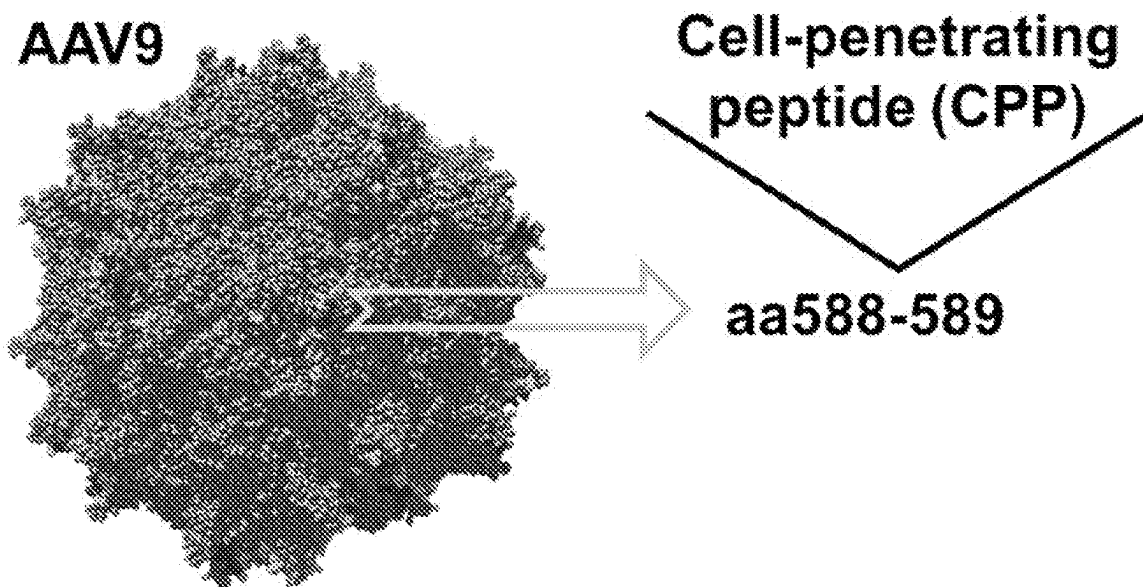


FIG. 1A

**Modification of AAV9
capsid plasmid**

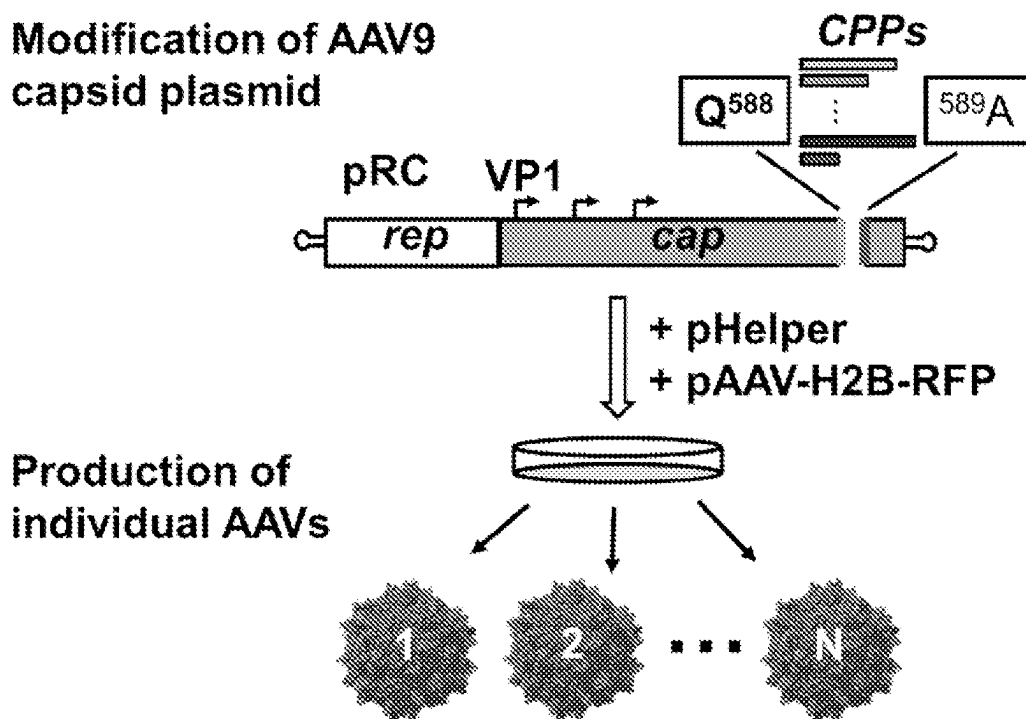


FIG. 1B

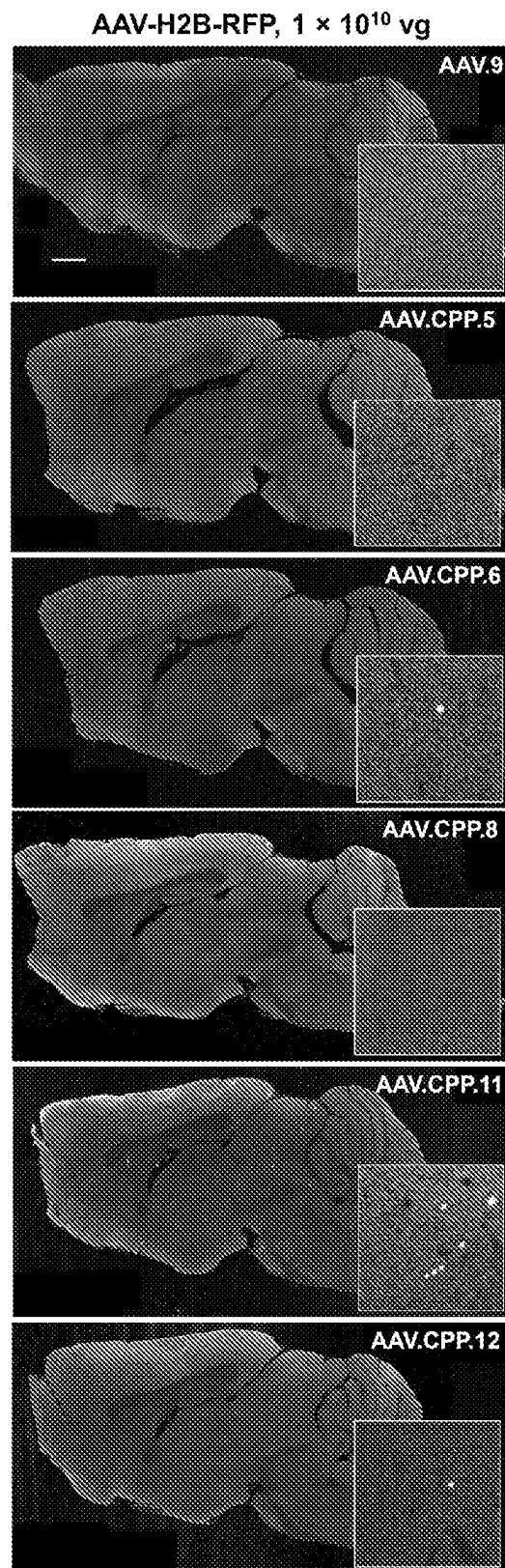


FIG. 2A

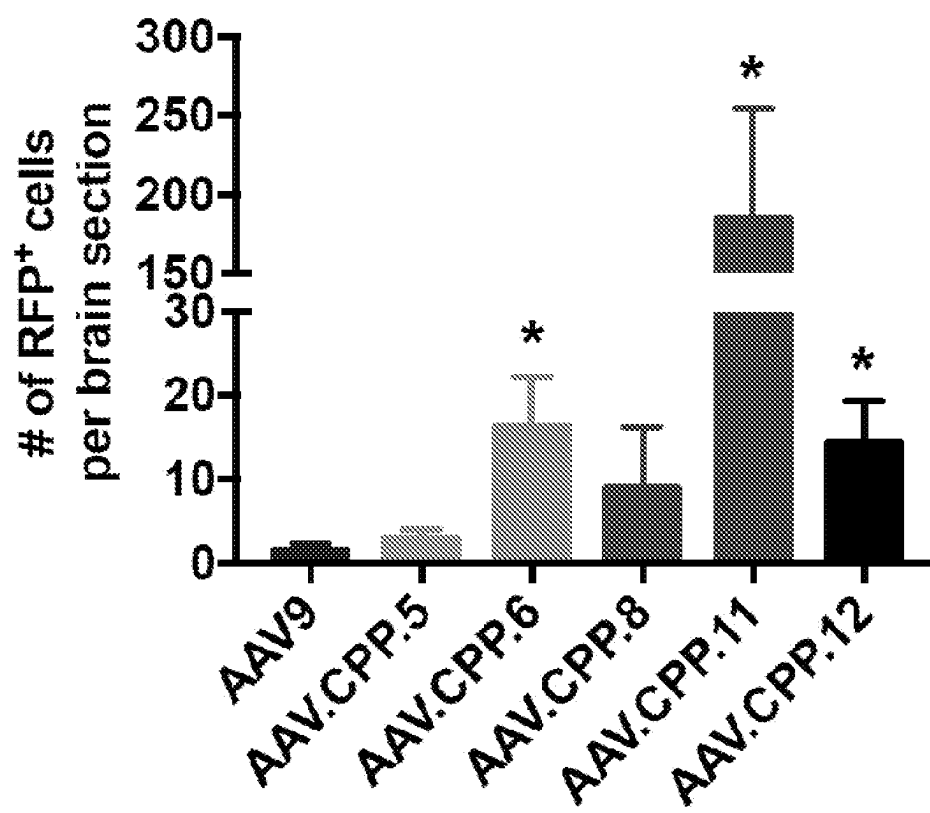


FIG. 2B

AAV-H2B-RFP, 1×10^{11} vg

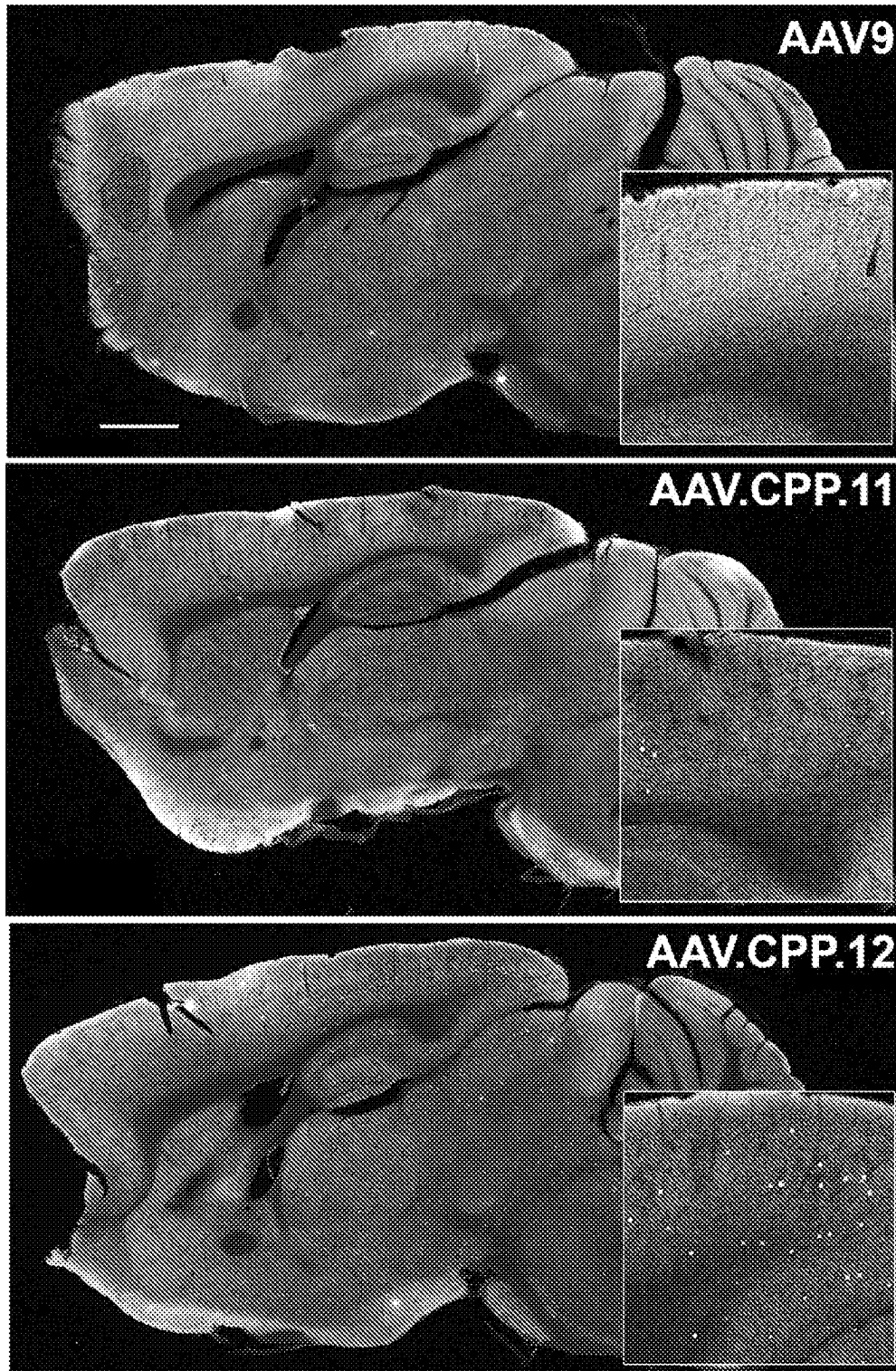
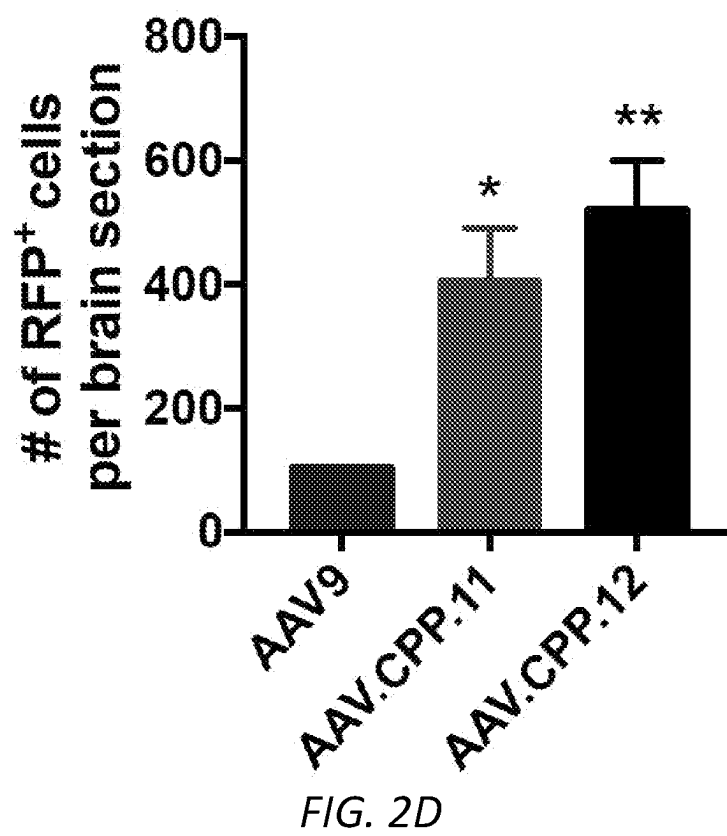


FIG. 2C



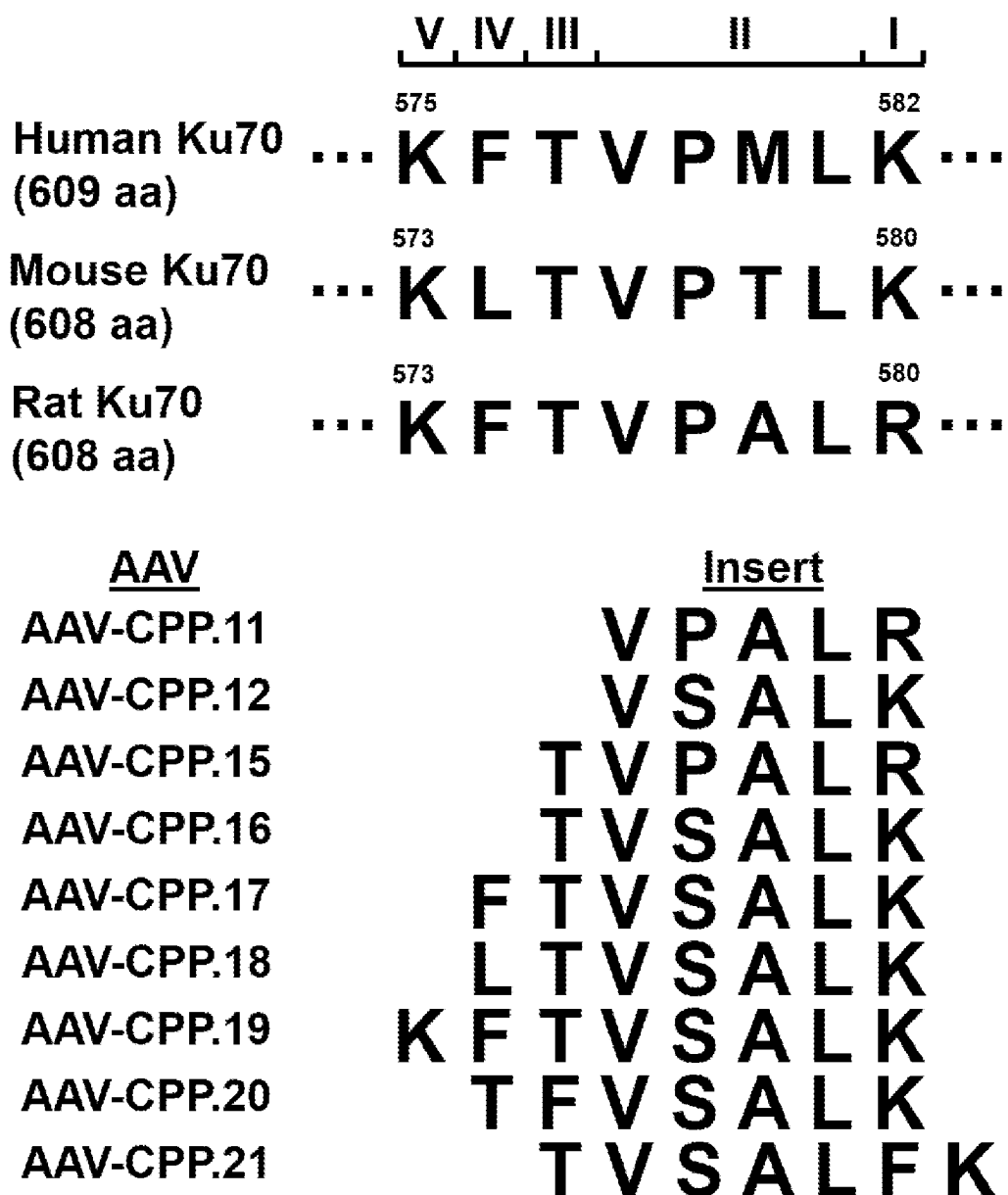
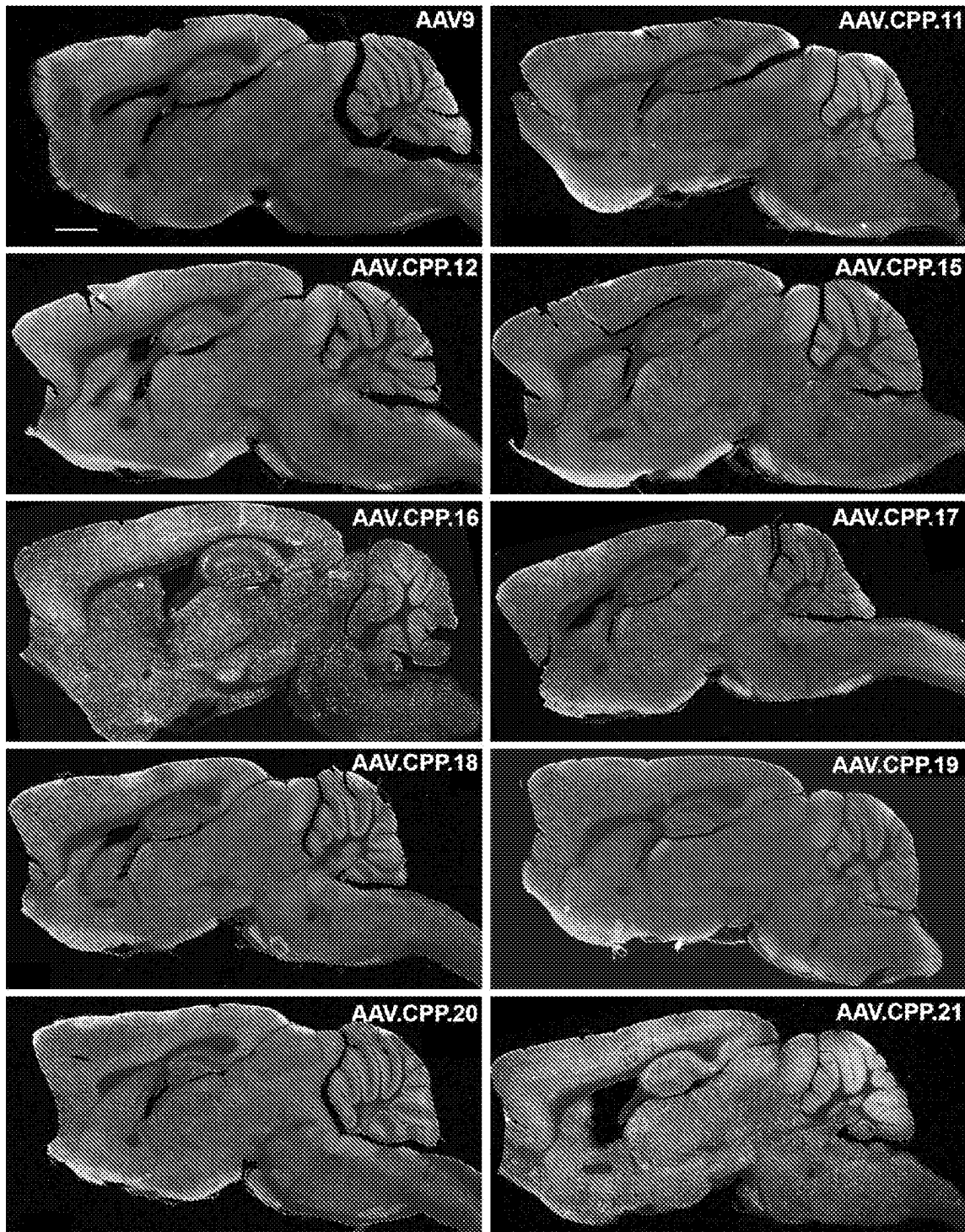


FIG. 3A

AAV-H2B-RFP, 1×10^{11} vg**FIG. 3B**

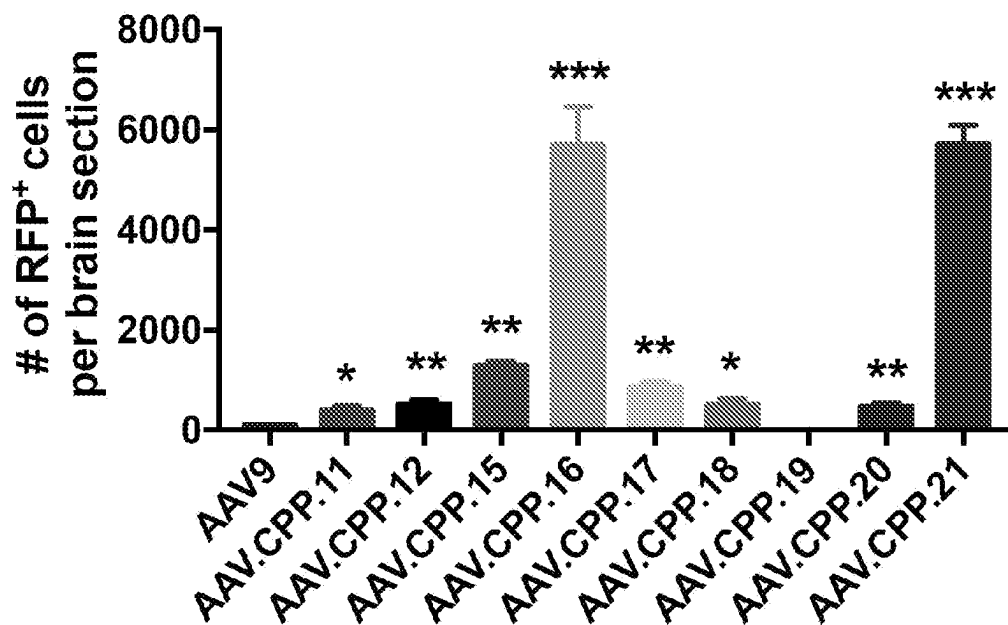


FIG. 3C

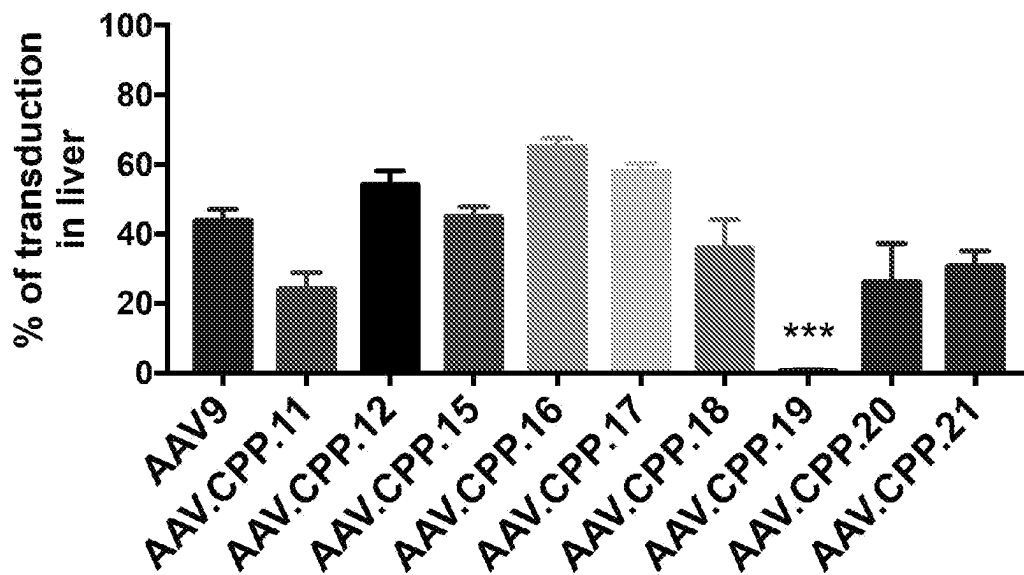
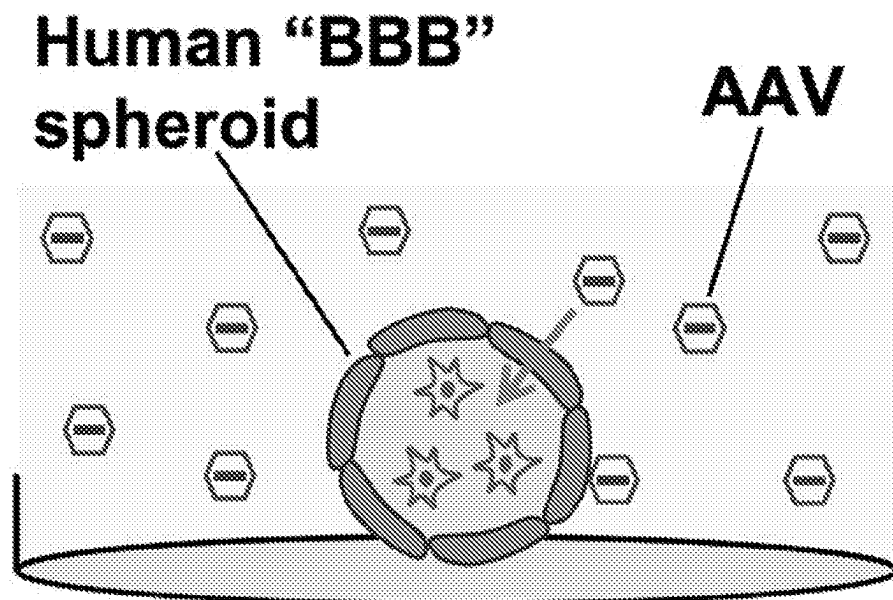
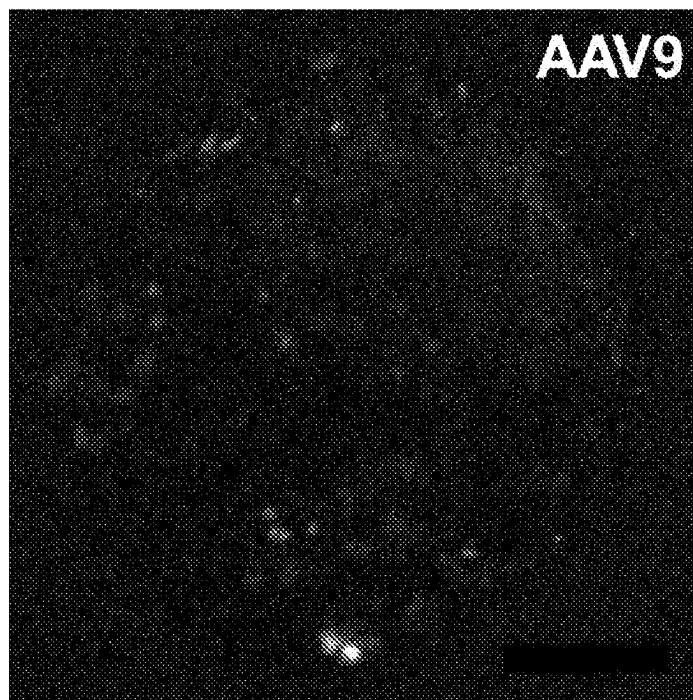


FIG. 3D

*FIG. 4A**FIG. 4B*

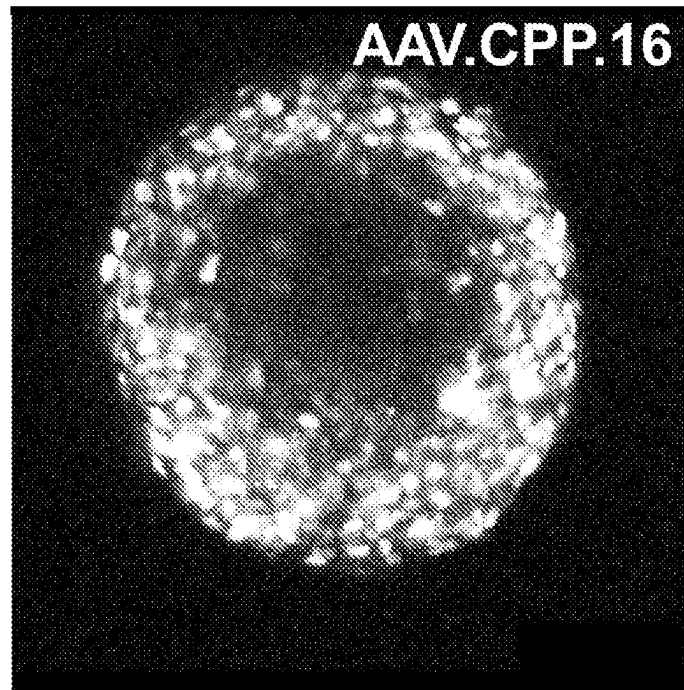


FIG. 4C

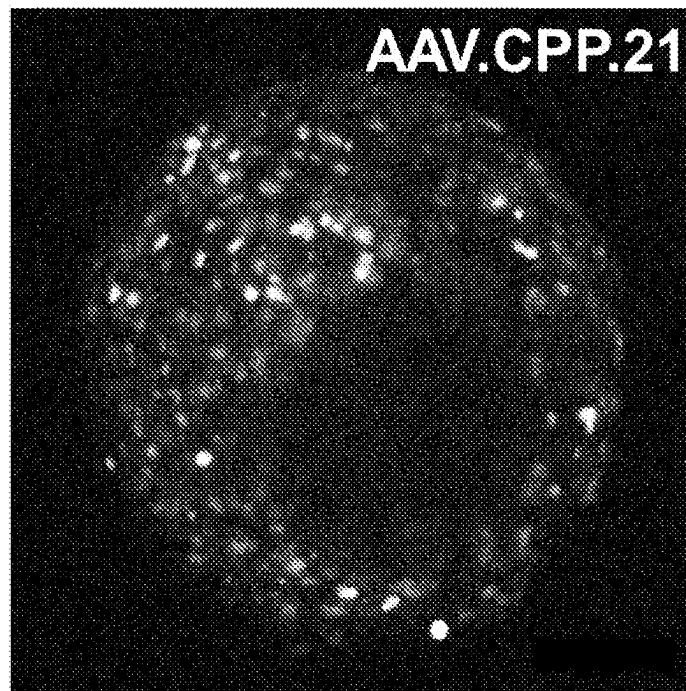


FIG. 4D

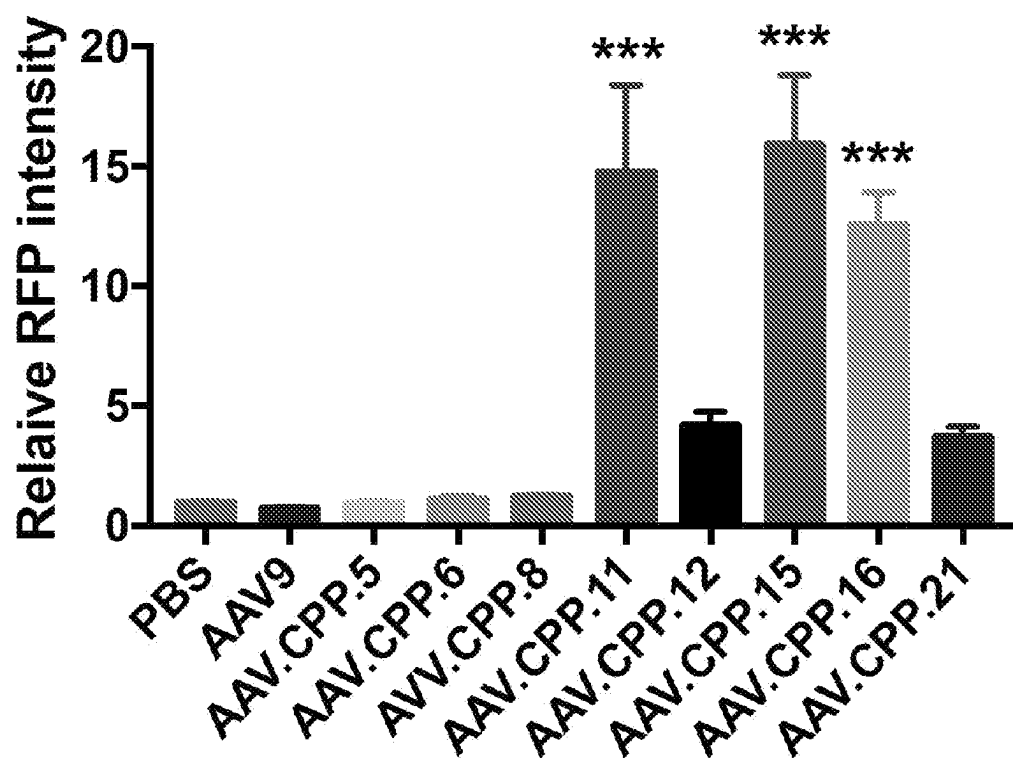


FIG. 4E

C57BL/6J, 1×10^{12} vg

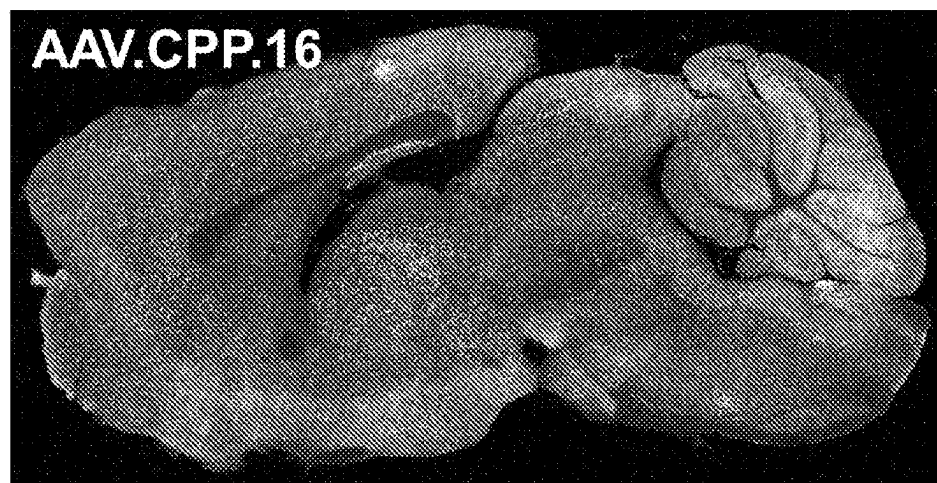
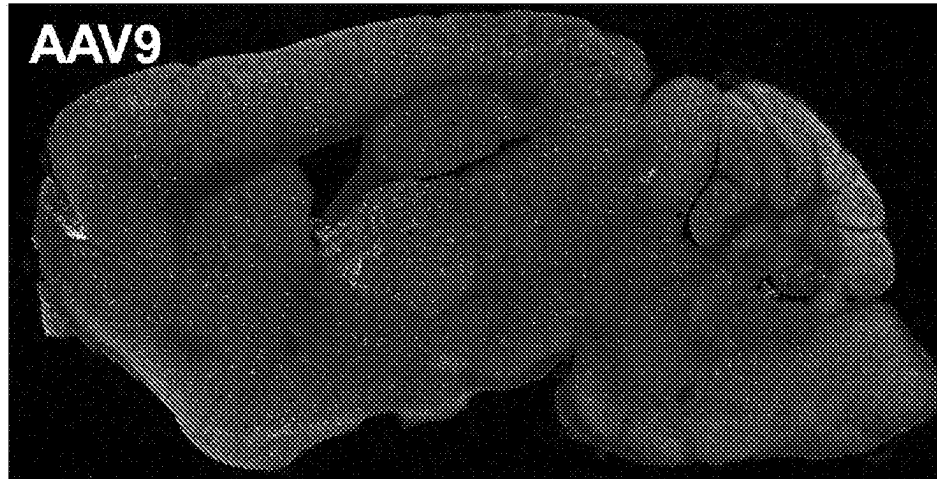


FIG. 5A

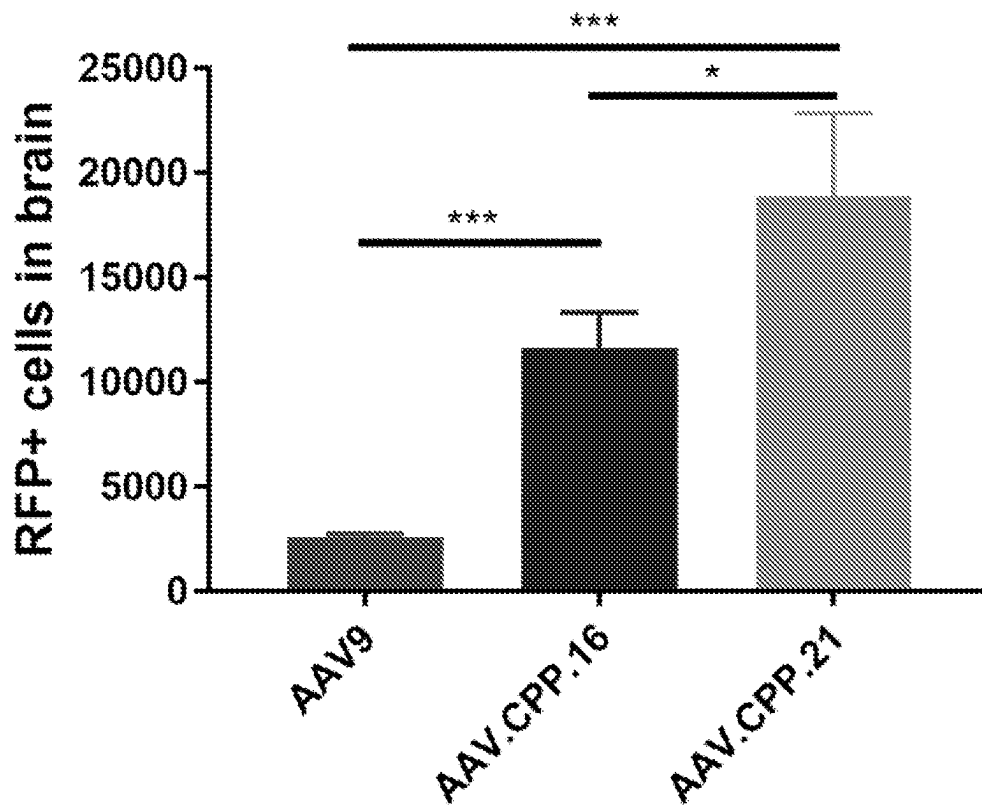


FIG. 5B

BALB/cJ, 1×10^{12} vg

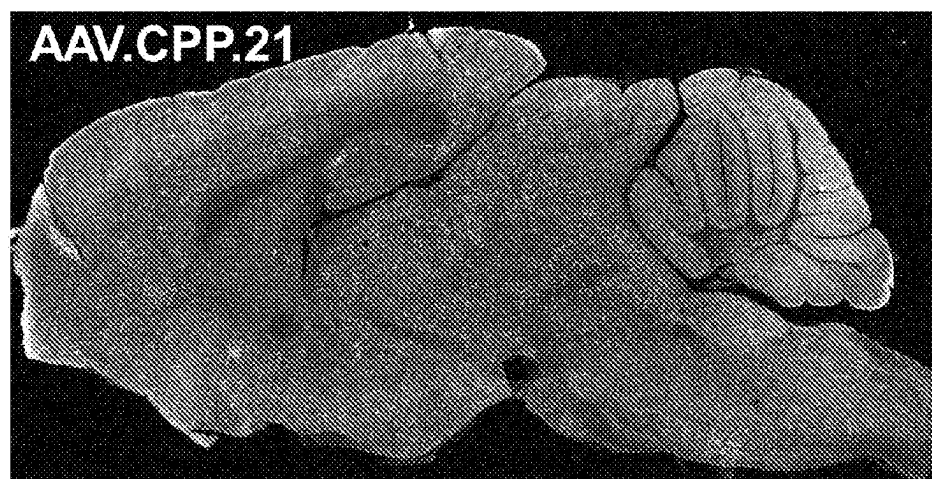
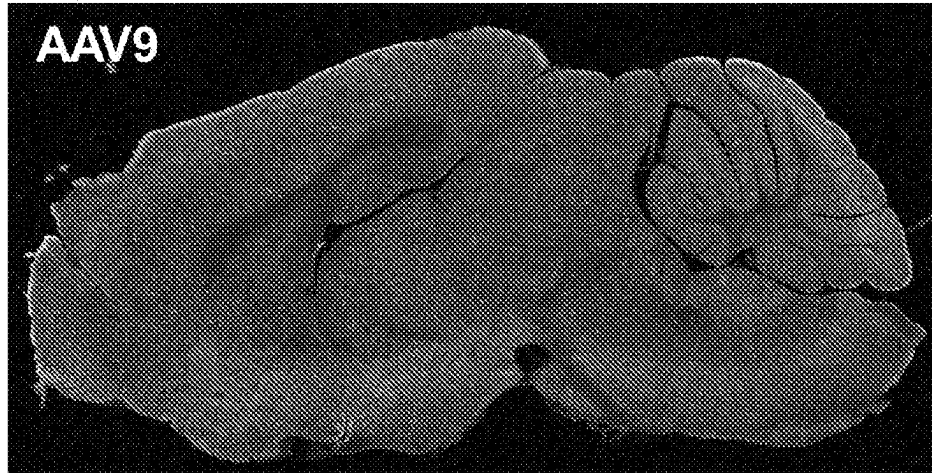


FIG. 6A

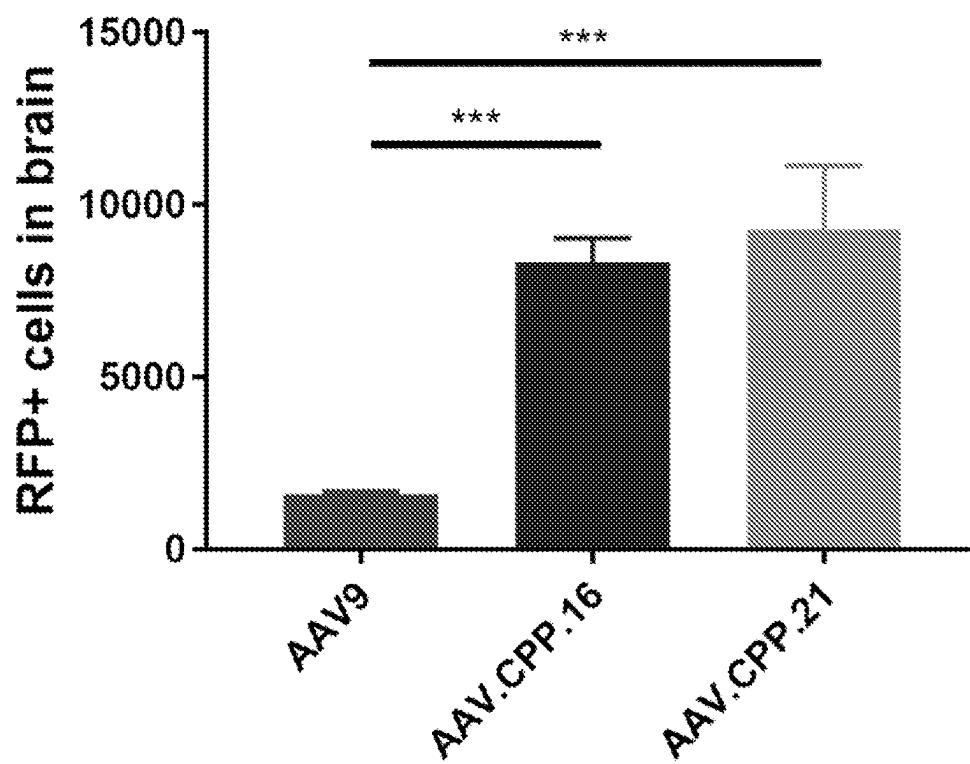
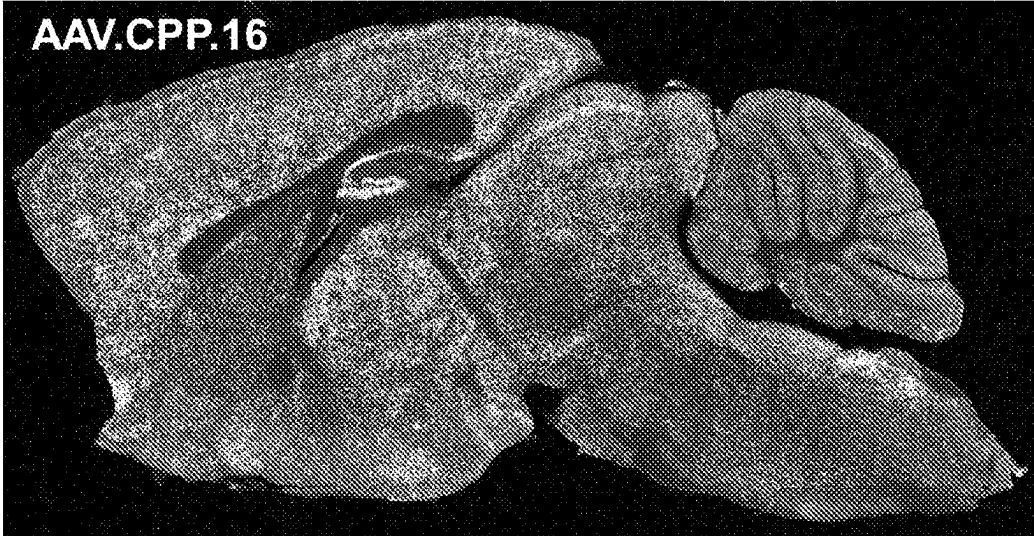


FIG. 6B

C57BL/6J, 4×10^{12} vg

AAV.CPP.16



AAV.CPP.21

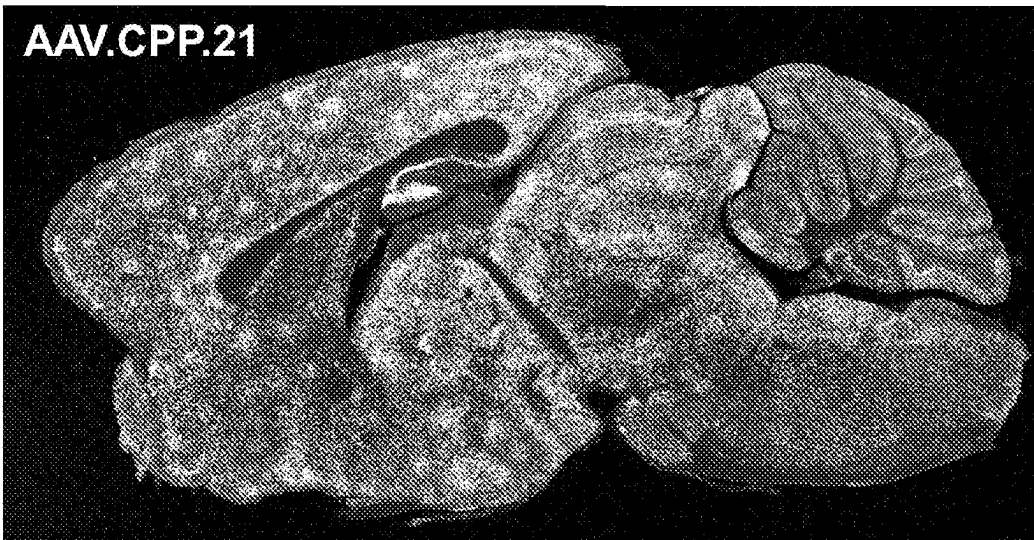


FIG. 7A

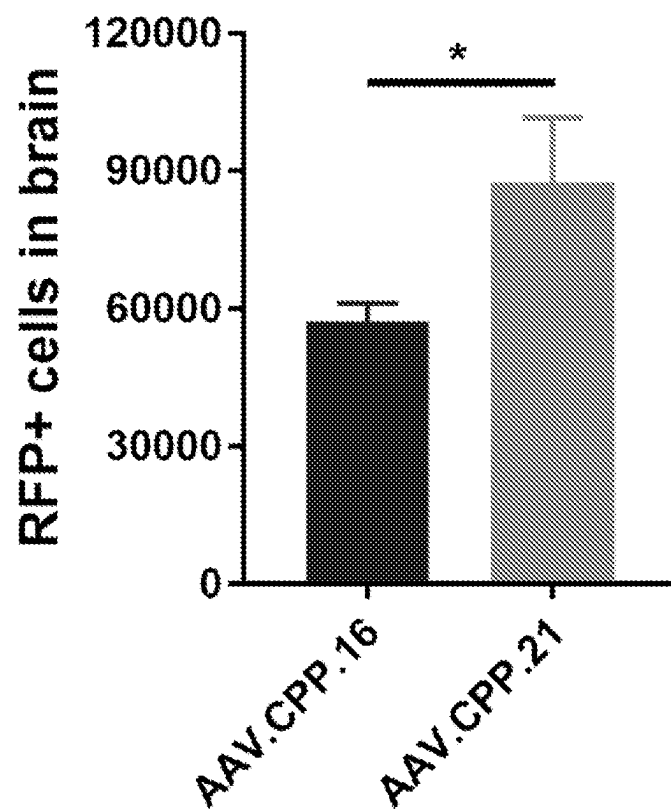
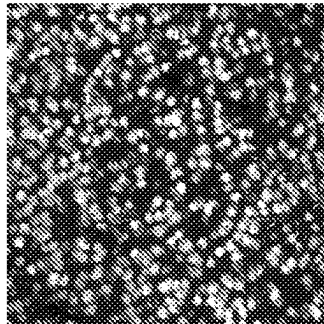
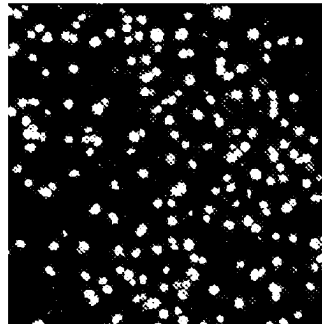
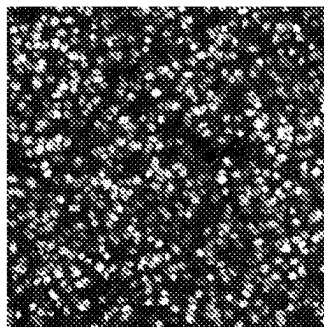
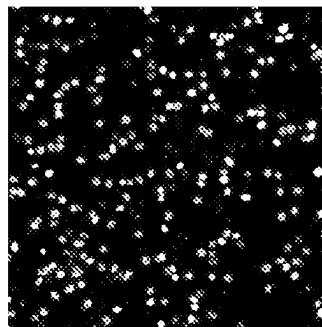
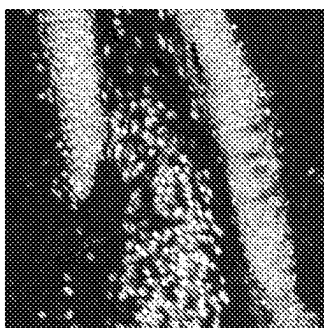
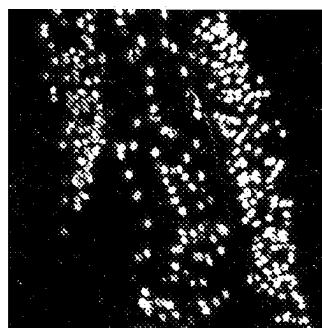
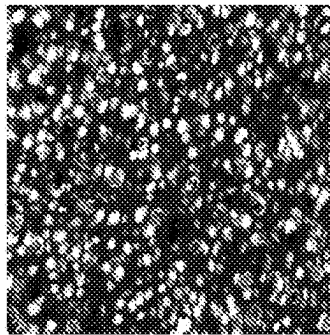
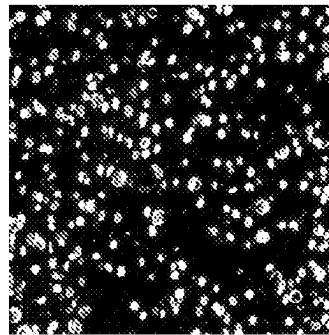
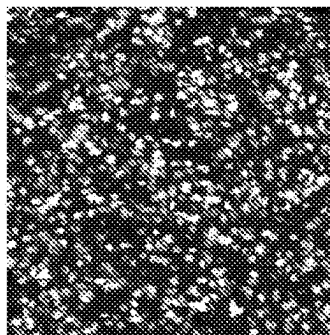
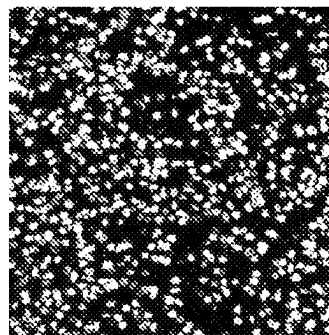
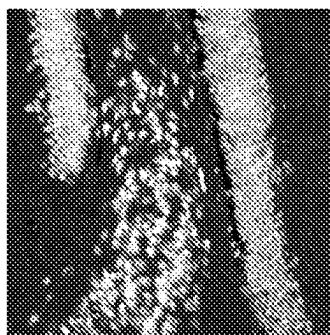
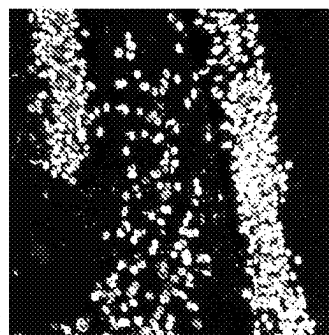


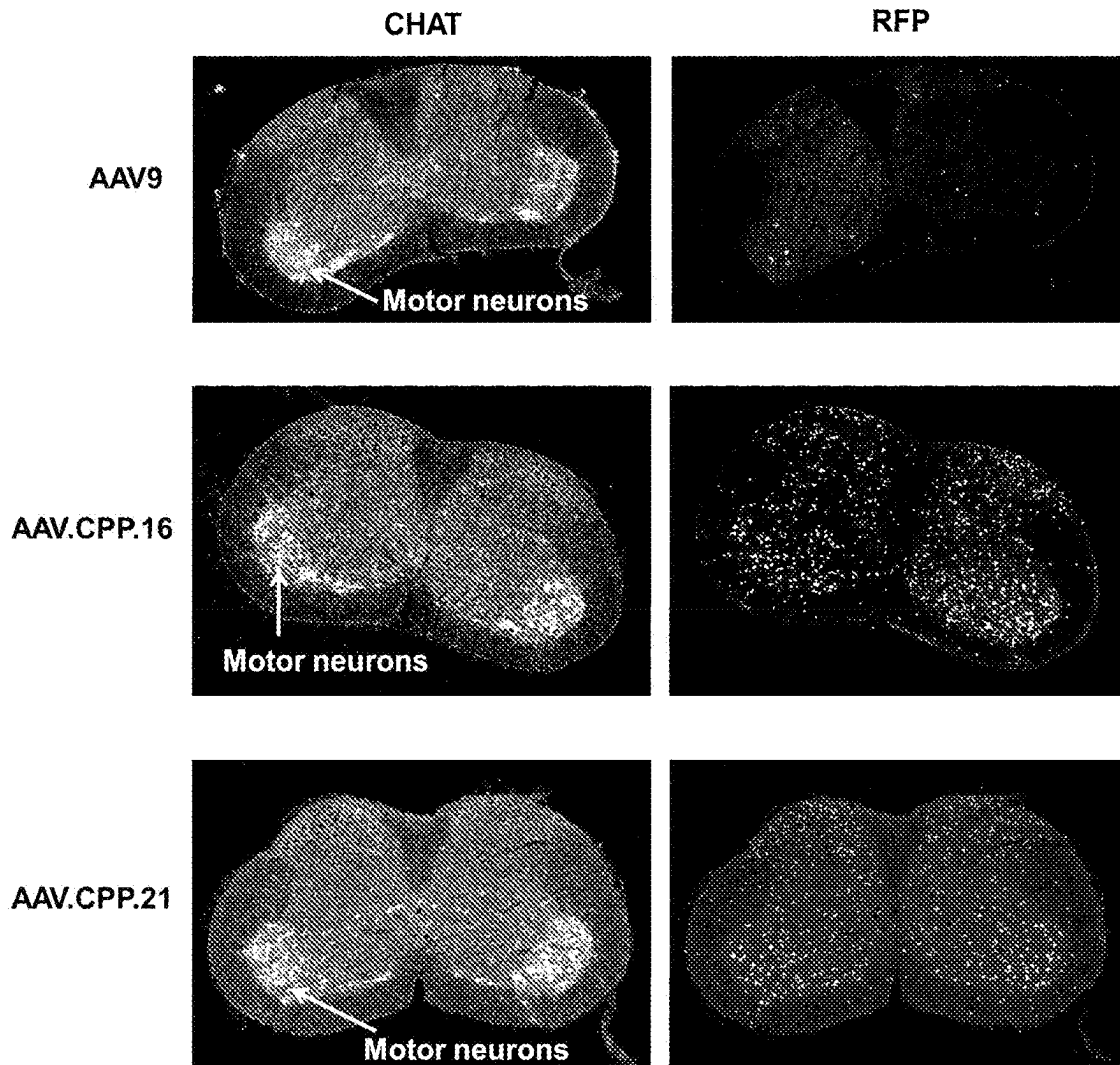
FIG. 7B

AAV.CPP.16

Cortex**NeuN****RFP****Midbrain****NeuN****RFP****Hippocampus****NeuN****RFP***FIG. 8A*

AAV.CPP. 21

Cortex**NeuN****RFP****Midbrain****NeuN****RFP****Hippocampus****NeuN****RFP***FIG. 8A, continued*

*FIG. 8B*

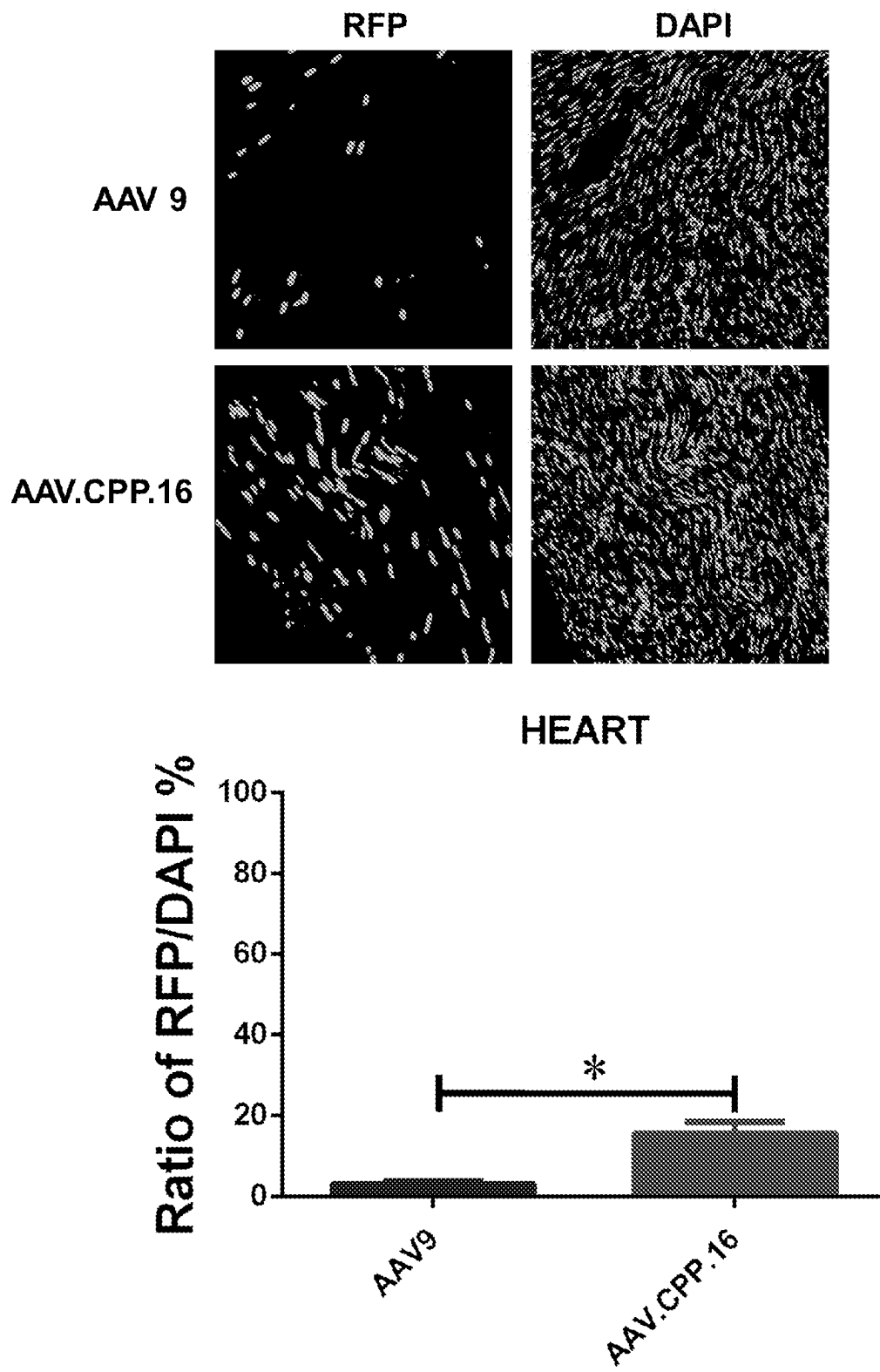


FIG. 9A

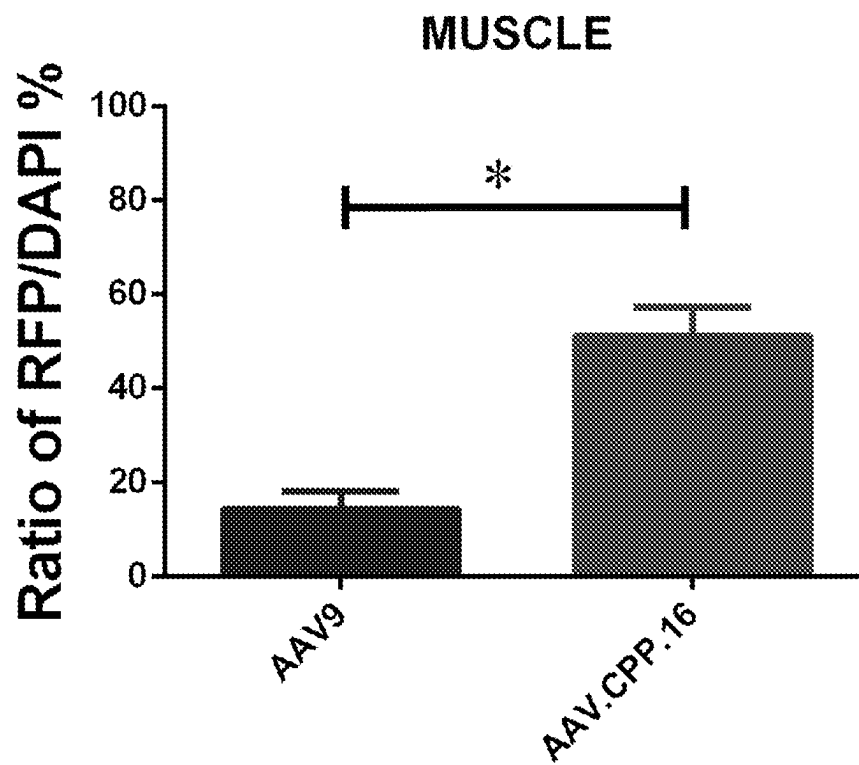
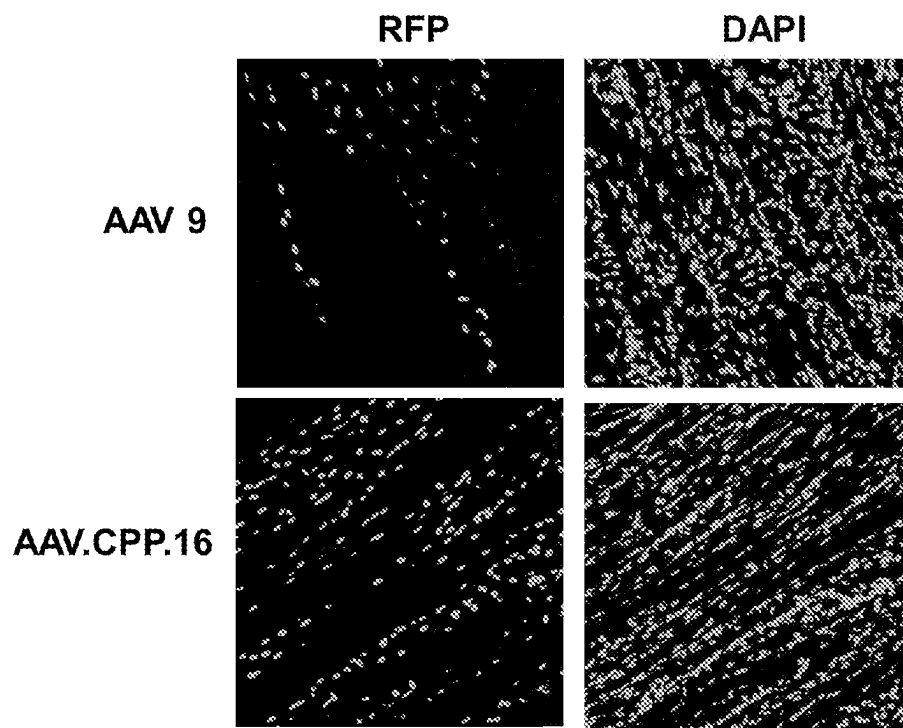


FIG. 9B

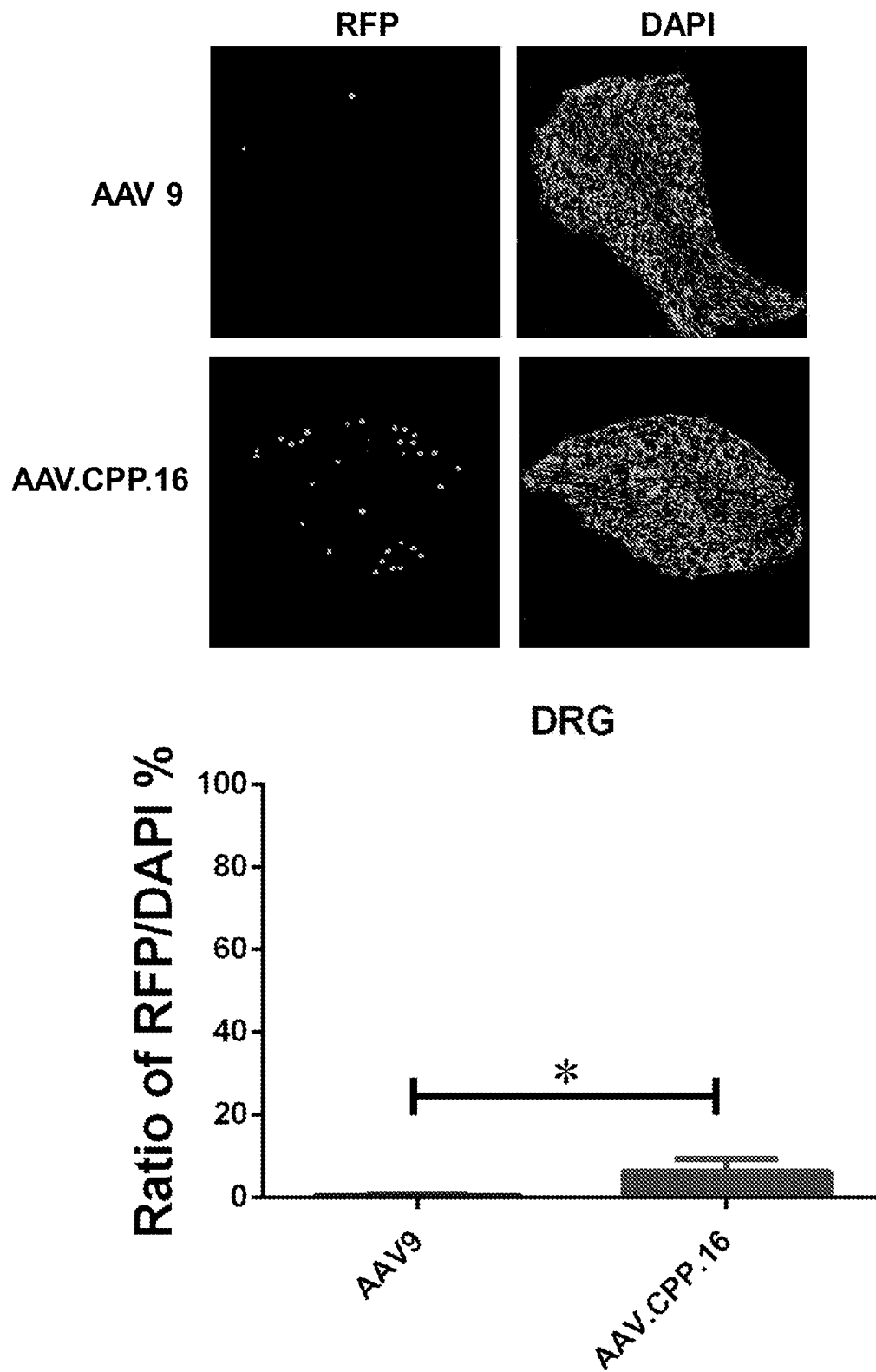
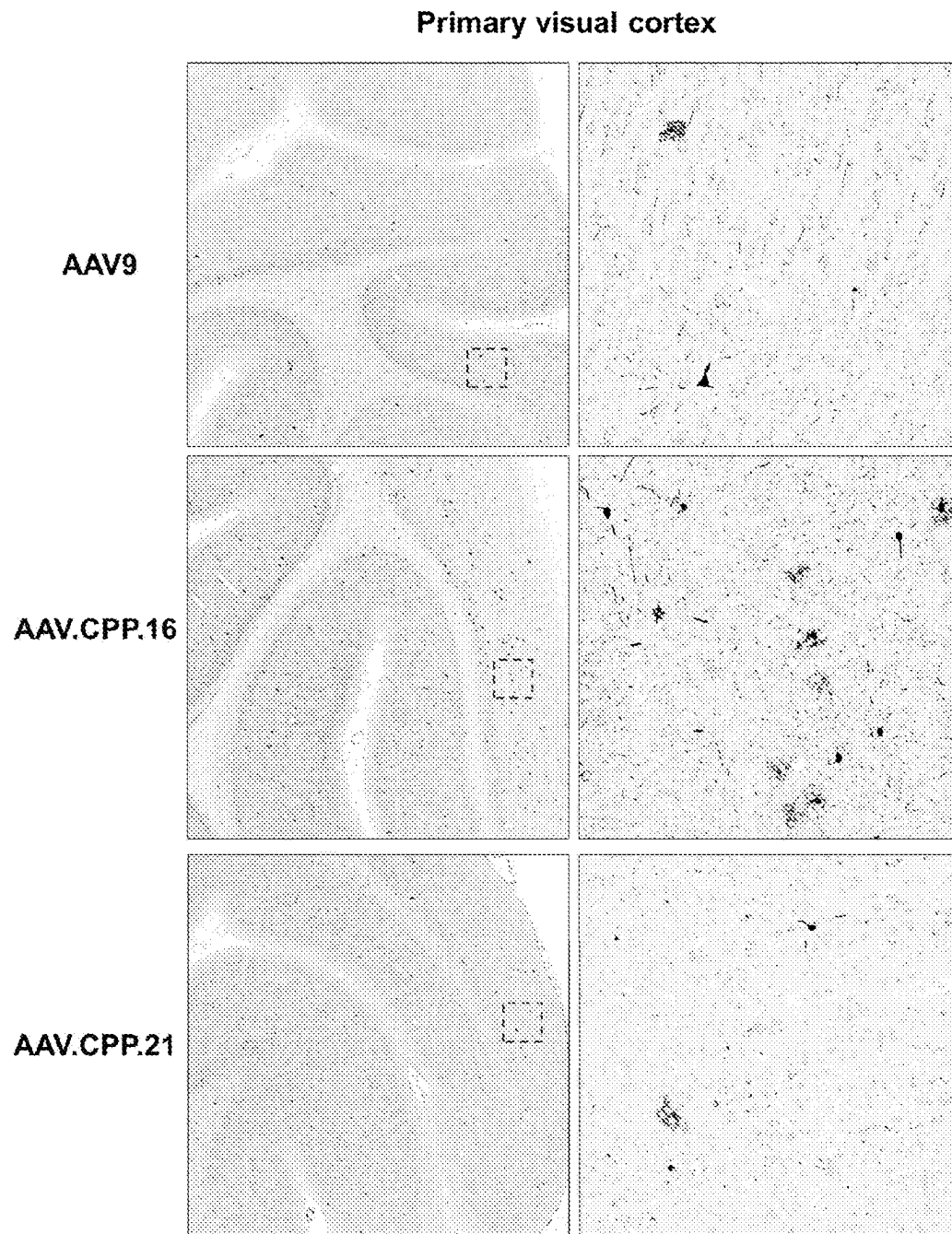


FIG. 9C

**FIG. 10A**

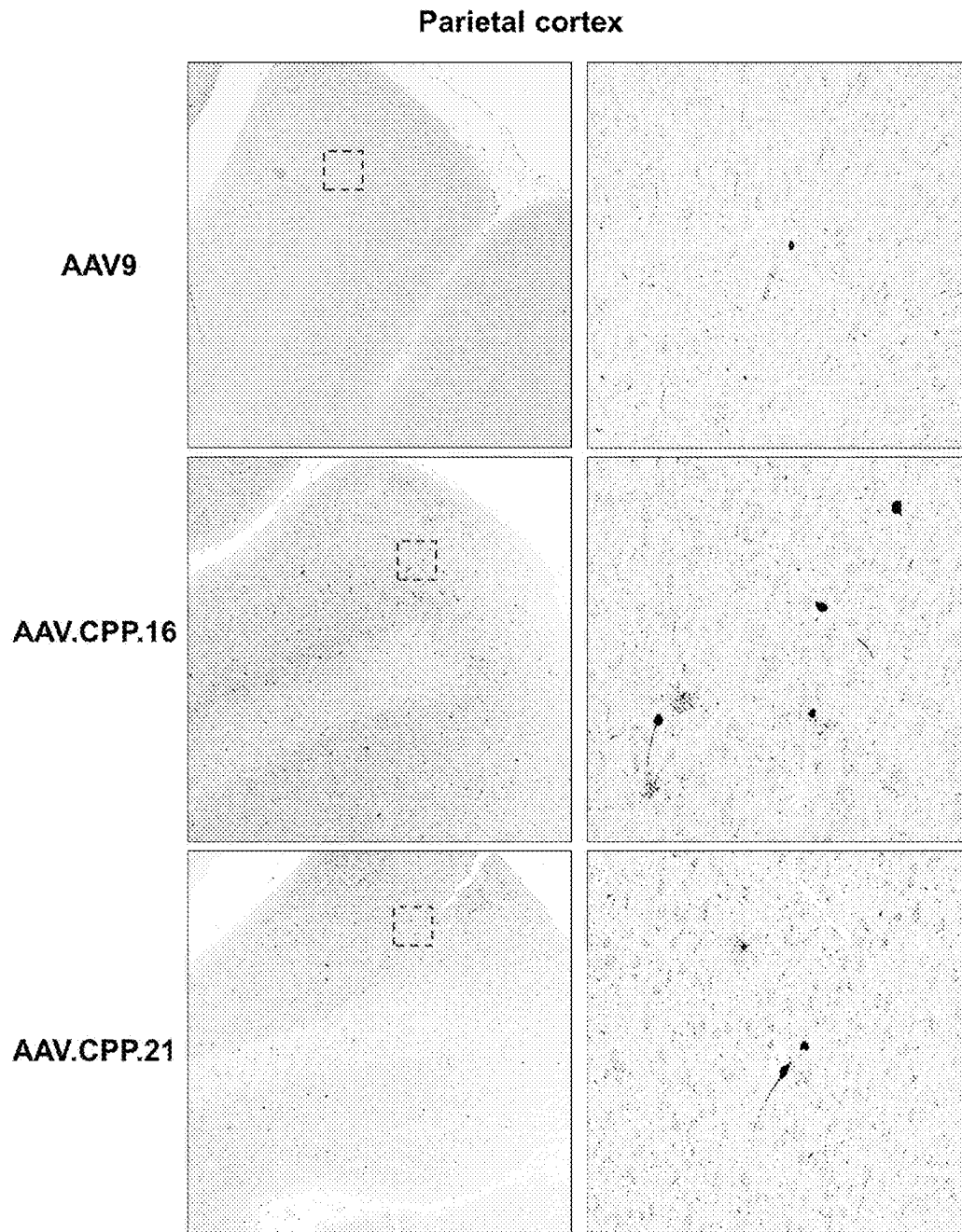


FIG. 10B

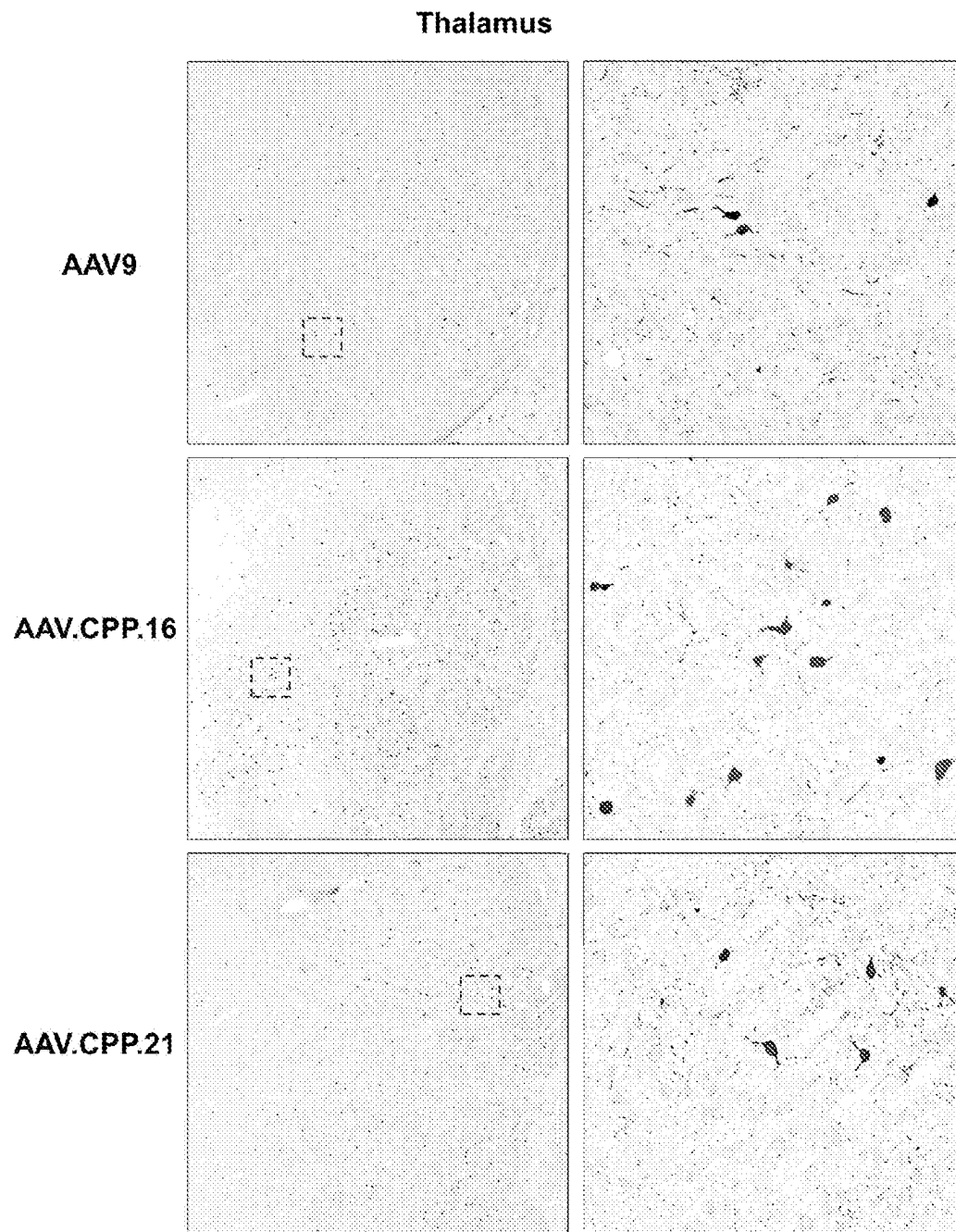
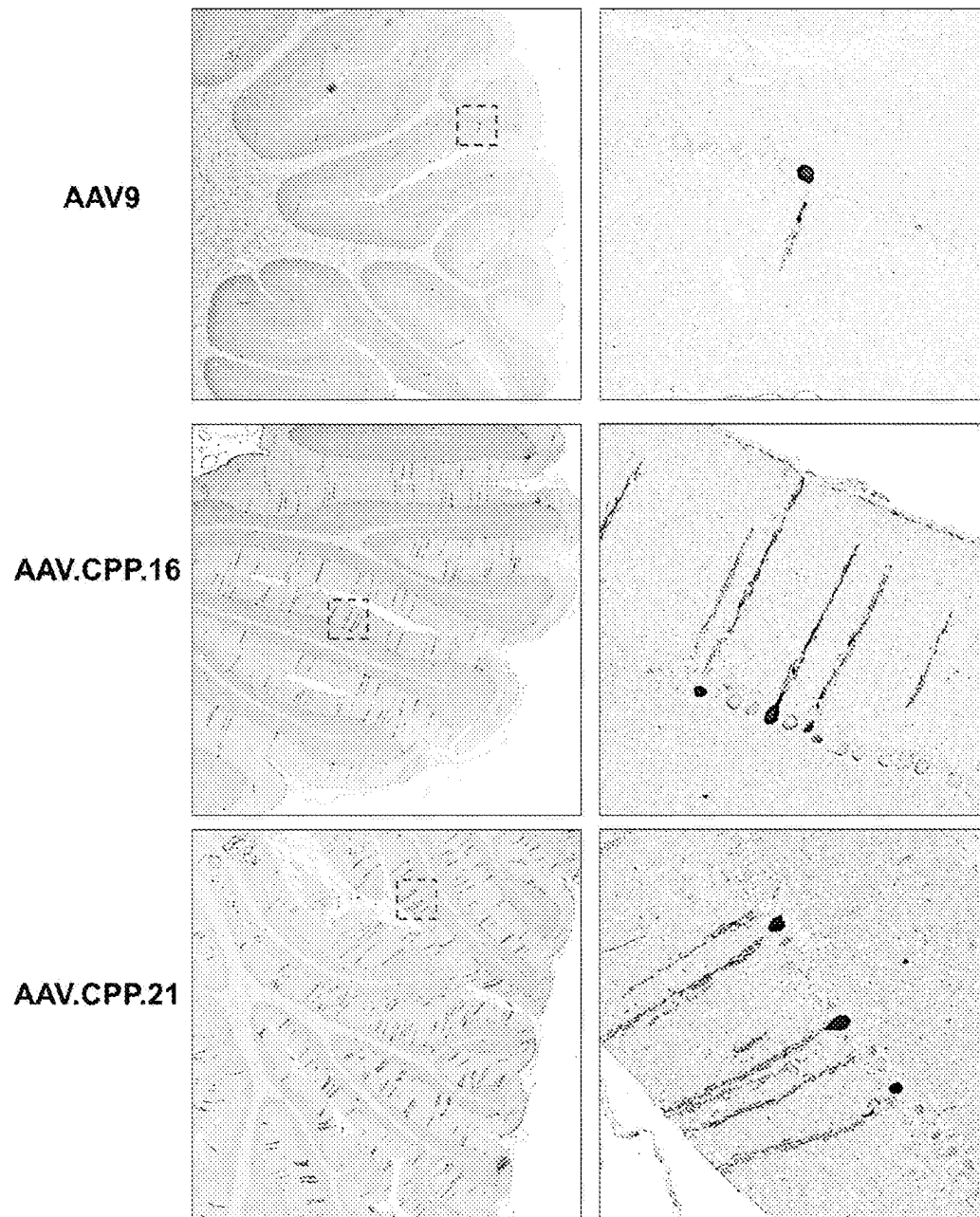


FIG. 10C

Cerebellum



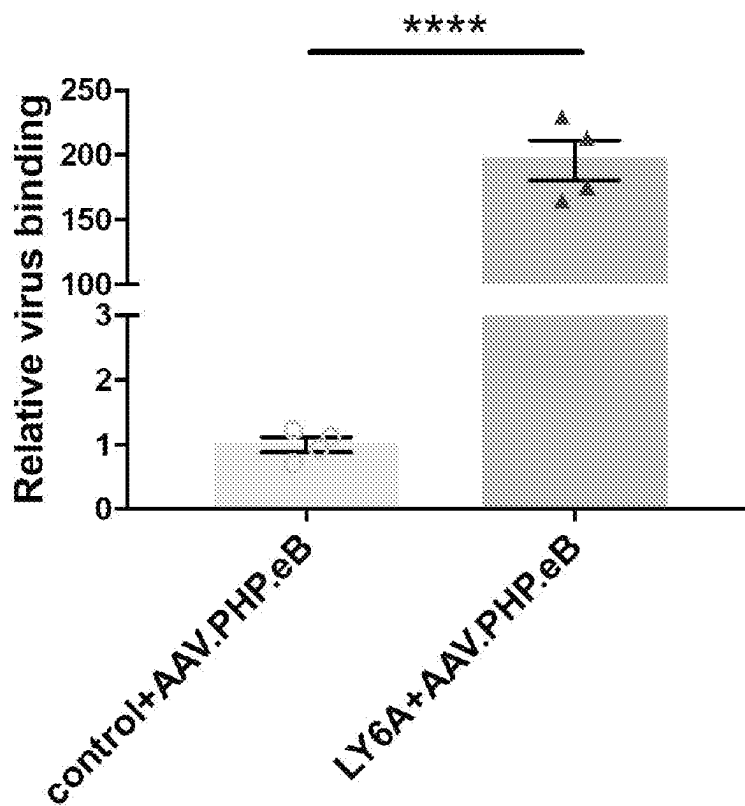


FIG. 11A

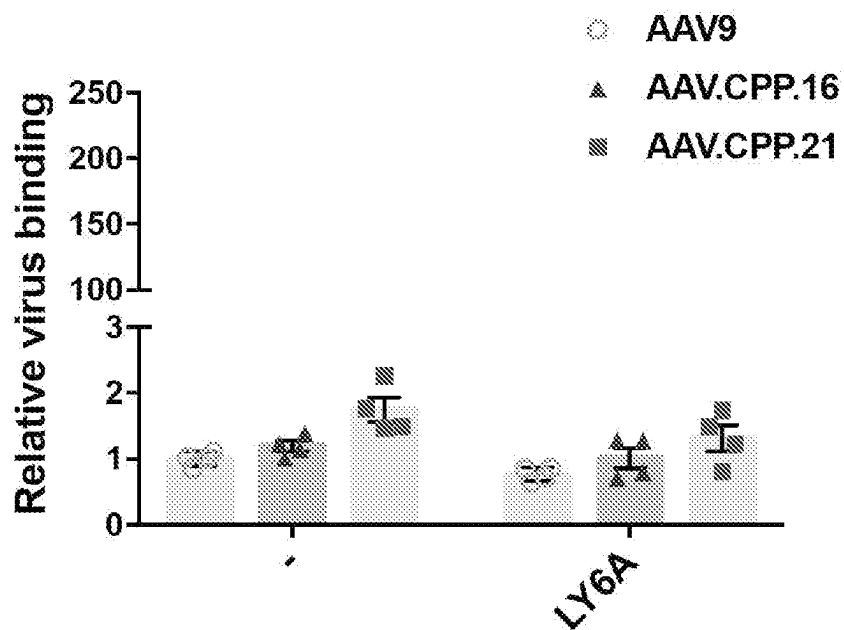
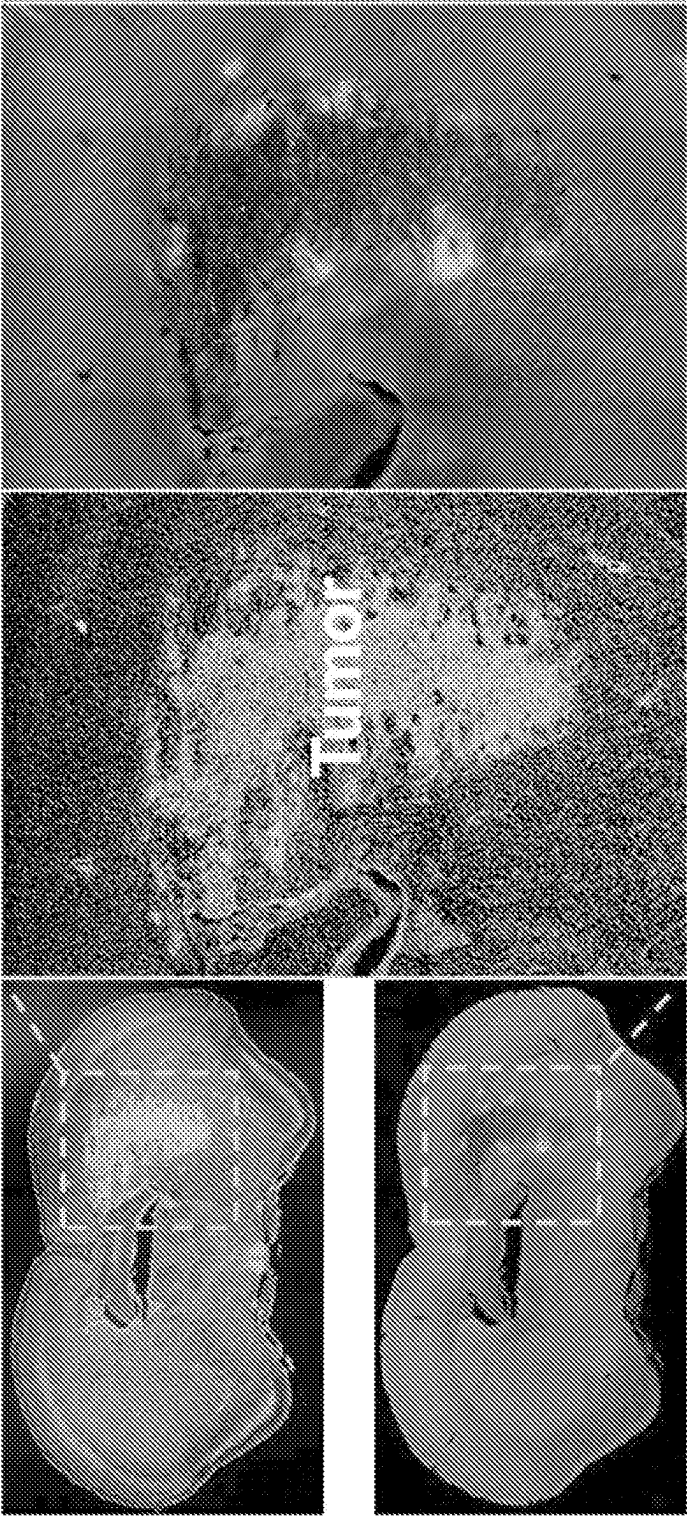


FIG. 11B

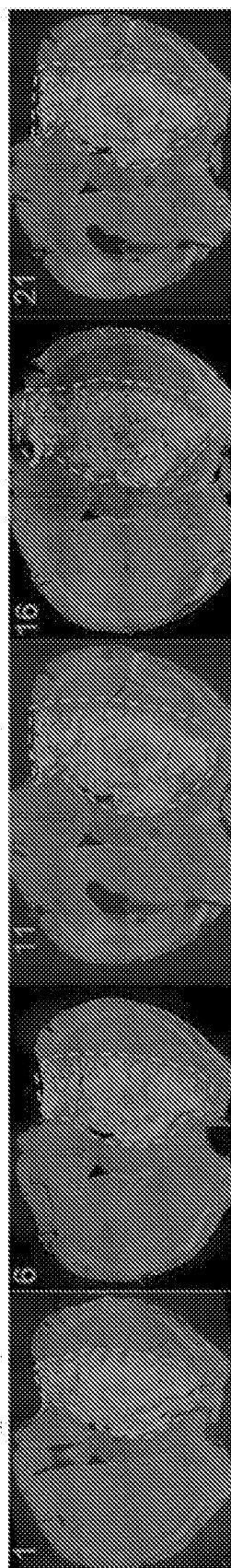
AAV-CPP.21-nestin-H2BmCherry, 1e11 vg; IV

DAPI mCherry



GL261 mouse tumor model FIG. 12A

Control group: AAV-CPP.21-nestin-HSV.TK1 + saline, IV



Drug group: AAV-CPP.21-nestin-HSV.TK1 + Ganciclovir, IV



FIG. 12B

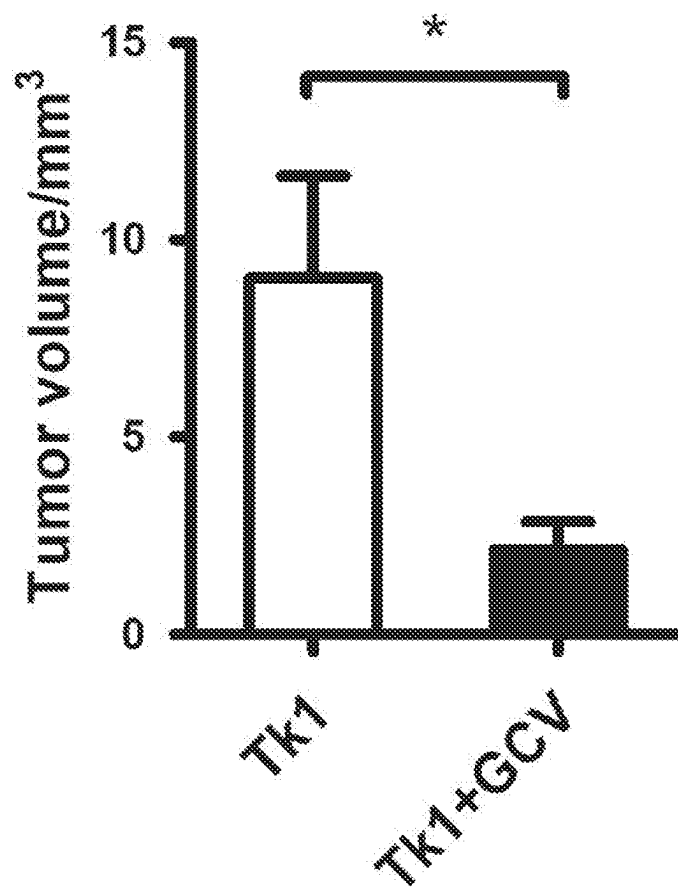


FIG. 12C

AAV9



AAV.CPP.21

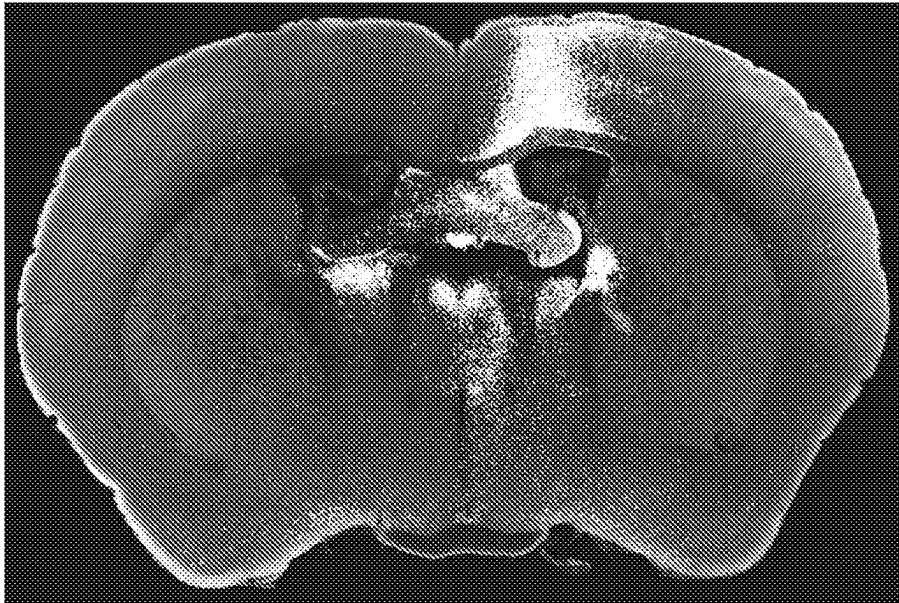


FIG. 13

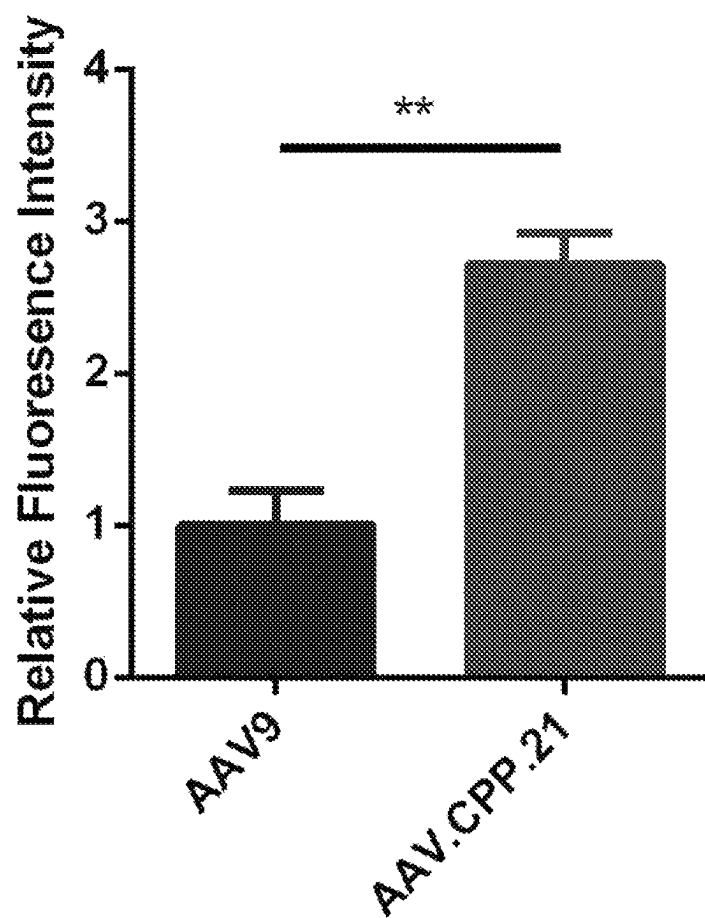


FIG. 13, continued

1

METHODS AND COMPOSITIONS FOR DELIVERY OF AGENTS ACROSS THE BLOOD-BRAIN BARRIER

CLAIM OF PRIORITY

This application is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/US2019/041386, having an International Filing Date of Jul. 11, 2019, which claims the benefit of U.S. Provisional Application Ser. No. 62/696,422, filed on Jul. 11, 2018. The entire contents of the foregoing are incorporated herein by reference.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Jul. 11, 2019, is named 29618-0200W01_SL.txt and is 58,834 bytes in size.

TECHNICAL FIELD

Described herein are sequences that enhance permeation of agents across the blood brain barrier, compositions comprising the sequences, and methods of use thereof.

BACKGROUND

Delivery of therapeutic agents, including gene therapy reagents, is an impediment to development of treatments for a number of conditions. The blood-brain barrier (BBB) is a key obstacle for drug delivery to the mammalian central nervous system (CNS), particularly for delivery to the human brain, to treat conditions including neurodegenerative diseases such as Parkinson's disease; Alzheimer's disease; Huntington's disease; Amyotrophic lateral sclerosis; and Multiple sclerosis.

SUMMARY

The present invention is based on the development of artificial targeting sequences that enhance permeation of agents into cells and across the blood brain barrier.

Thus provided herein is an AAV capsid protein, e.g., an engineered AAV capsid protein, comprising a targeting sequence that comprises at least four contiguous amino acids from the sequence TVSALFK (SEQ ID NO:8); TVSALK (SEQ ID NO:4); KLASVT (SEQ ID NO:83); or KFLASVT (SEQ ID NO:84). In some embodiments, the AAV capsid protein comprises a targeting sequence that comprises at least five contiguous amino acids from the sequence TVSALK (SEQ ID NO:4); TVSALFK (SEQ ID NO:8); KLASVT (SEQ ID NO:83); or KFLASVT (SEQ ID NO:84). In some embodiments, the AAV capsid protein comprises a targeting sequence that comprises at least six contiguous amino acids from the sequence TVSALK (SEQ ID NO:4); TVSALFK (SEQ ID NO:8); KLASVT (SEQ ID NO:83); or KFLASVT (SEQ ID NO:84).

In some embodiments, the AAV is AAV9; other AAV as known in the art (e.g., AAV1, 2, 3, 4, 5, 6, 7, 8 and variants thereof and others as known in the art or described herein) can also be used.

In some embodiments, the AAV capsid protein comprises AAV9 VP1 (e.g., SEQ ID NO:85).

2

In some embodiments, the targeting sequence is inserted in the capsid protein at a position corresponding to between amino acids 588 and 589 of SEQ ID NO:85.

Also provided herein are nucleic acids encoding the AAV capsid proteins comprising a targeting sequence as described herein.

In addition, provided herein is an AAV comprising a capsid protein comprising a targeting sequence as described herein. In some embodiments, AAV further comprises a transgene, preferably a therapeutic or diagnostic transgene. Therapeutic transgenes can include, e.g., cDNAs that restore protein function, guide RNA for gene editing, RNA, or miRNA.

Also provided herein are targeting sequences comprising V[S/p][A/m/t]/L (SEQ ID NO:79), TV[S/p][A/m/t]/L (SEQ ID NO:80), TV[S/p][A/m/t]/LK (SEQ ID NO:81), or TV[S/p][A/m/t]/LFK (SEQ ID NO:82). In some embodiments, the targeting sequence comprises VPALR (SEQ ID NO: 1); VSALK (SEQ ID NO:2); TVPALR (SEQ ID NO:3); TVSALK (SEQ ID NO:4); TVPMLK (SEQ ID NO: 12); TVPTLK (SEQ ID NO:13); FTVSALK (SEQ ID NO:5); LTVSALK (SEQ ID NO:6); TVSALFK (SEQ ID NO:8); TVPALFR (SEQ ID NO:9); TVPMLFK (SEQ ID NO:10) or TVPTLFK (SEQ ID NO:11). Also provided are fusion proteins comprising the targeting sequences linked to a heterologous (e.g., non-AAV VP1) sequence, and AAV capsid proteins (e.g., AAV9 VP1) comprising the targeting sequence. In some embodiments, the targeting sequence is inserted in a position corresponding to amino acids 588 and 589 of SEQ ID NO:85.

Additionally provided herein are nucleic acids encoding the targeting sequences, fusion proteins or AAV capsid proteins described herein, as well as AAV comprising the capsid proteins comprising a targeting sequence. In some embodiments, the AAV further comprises a transgene, preferably a therapeutic or diagnostic transgene. Therapeutic transgenes can include, e.g., cDNAs that restore protein function, guide RNA for gene editing, RNA, or miRNA.

Further, provided herein are methods of delivering a transgene to a cell, the method comprising contacting the cell with an AAV or fusion protein described herein. In some embodiments, the cell is in a living subject, e.g., a mammalian subject. In some embodiments, the cell is in a tissue selected from the brain, spinal cord, dorsal root ganglion, heart, or muscle, and a combination thereof. In some embodiments, the cell is a neuron (optionally a dorsal root ganglion neuron), astrocyte, cardiomyocyte, or myocyte.

In some embodiments, the subject has a neurodegenerative disease, epilepsy; stroke; spinocerebellar ataxia; Canavan's disease; Metachromatic leukodystrophy; Spinal muscular atrophy; Friedreich's ataxia; X-linked centronuclear myopathy; Lysosomal storage disease; Barth syndrome; Duchenne muscular dystrophy; Wilson's disease; or Crigler-Najjar syndrome type 1. In some embodiments, the neurodegenerative disease is Parkinson's disease; Alzheimer's disease; Huntington's disease; Amyotrophic lateral sclerosis; and Multiple sclerosis.

In some embodiments, the subject has a brain cancer, and the method includes administering an AAV encoding an anti-cancer agent. In some embodiments, the anti-cancer agent is HSV.TK1, and the method further comprises administering ganciclovir.

In some embodiments, the cell is in the brain of the subject, and the AAV is administered by parenteral delivery (e.g., via intravenous, intraarterial, subcutaneous, intraperitoneal, or intramuscular delivery); intracerebral; or intrath-

ecal delivery (e.g., via lumbar injection, cisternal *magna* injection, or intraparenchymal injection).

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Methods and materials are described herein for use in the present invention; other, suitable methods and materials known in the art can also be used. The materials, methods, and examples are illustrative only and not intended to be limiting. All publications, patent applications, patents, sequences, database entries, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control.

Other features and advantages of the invention will be apparent from the following detailed description and figures, and from the claims.

DESCRIPTION OF DRAWINGS

FIGS. 1A-1B depict an exemplary strategy of engineering AAV9 by inserting cell-penetrating peptides (CPPs) into its capsid. FIG. 1A is a 3D model of an AAV9 virus. Individual CPP inserted into the capsid between amino acids 588 and 589 (VP1 numbering) will be displayed at the 3-fold axis where receptor binding presumably occurs. FIG. 1B illustrates the method of individual AAV production. Three plasmids including pRC (engineered or not), pHelper and pAAV are co-transfected into HEK 293T cells, with AAVs harvested and purified using iodixanol gradient.

FIGS. 2A-2B depict representative images of mouse brain sections and their quantitative analysis after intravenous administration of low-dose candidate AAVs. Mice with mixed genetic background are used. Candidate AAVs differs in their inserted CPPs (see Table 3), but all express nuclear red fluorescent protein (RFP) as reporter. Candidate AAVs with low production yields are excluded for further screening. The dose of AAV is 1×10^{10} vg (viral genome) per animal. Each white dot in FIG. 2A represents a RFP-labeled cell. In FIG. 2B, * $P < 0.05$, vs. AAV9, ANOVA.

FIGS. 2C-2D depict representative images of mouse brain sections and their quantitative analysis after intravenous administration of AAV.CPP.11 and AAV.CPP.12 in a repeat experiment. AAV.CPP.11 and AAV.CPP.12 contain CPPs BIP1 and BIP2 respectively (see Table 3). The doses of the AAVs are increased to 1×10^{11} vg per animal. Candidate AAVs express nuclear red fluorescent protein (RFP) as reporter. Each white dot in FIG. 2C represents a RFP-labeled cell. In FIG. 2D, * $P < 0.05$, ** $P < 0.01$, vs. AAV9, ANOVA.

FIG. 3A depicts the optimization of the BIP targeting sequence in order to further engineer AAV9 towards better brain transduction. BIP1 (VPALR, SEQ ID NO:1), which enables AAV9 to transduce brain more efficiently (as in AAV.CPP.11), is derived from the protein Ku70 in rats. Human, mouse and rat Ku70 proteins differ in their exact amino acid sequences. BIP2 (VSALK, SEQ ID NO:2) as in AAV.CPP.12 is a "synthetic" peptide related to BIP1. Further engineering focuses on the VSALK sequence in the hope of minimizing species specificity of final engineered AAV. To generate new targeting sequence, amino acids of interest are added to the VSALK sequence, and in other cases positions of individual amino acids are switched. All new BIP2-derived sequences are again inserted into the AAV9 capsid to generate new candidate AAVs for screening. Sequences appearing in order are SEQ ID NOs: 69, 70, 71, 1-6, 72, 7, and 8.

FIGS. 3B-3C depict representative images of mouse brain sections and their quantitative analysis after intravenous administration of more candidate AAVs. All candidate AAVs express nuclear red fluorescent protein (RFP) as reporter. The dose of AAV is 1×10^{11} vg per animal. Each white dot in FIG. 3B represents a RFP-labeled cell. AAV.CPP.16 and AAV.CPP.21 were identified as top hits with their robust and widespread brain transduction. In FIG. 3C, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, vs. AAV9, ANOVA.

FIG. 3D depicts quantitative analysis of transduction efficiency in the liver after intravenous administration of candidate AAVs. Percentage of transduced liver cells is presented. The dose of AAV is 1×10^{11} vg per animal. *** $P < 0.001$, vs. AAV9, ANOVA.

FIGS. 4A-4E depict screening of selected candidate AAVs in an in vitro spheroid model of human blood-brain barrier. FIG. 4A illustrates the spheroid comprising human microvascular endothelial cells, which forms a barrier at the surface, and human pericyte and astrocytes inside the spheroid. Candidate AAVs were assessed for their ability to penetrate from the surrounding medium into the inside of the spheroid and to transduce the cells inside. FIG. 4B-4D shows images of AAV9, AAV.CPP.16 and AAV.CPP.21 treated spheroids. FIG. 4E shows relative RFP intensity of different AAV treated spheroids. *** $P < 0.001$, vs. AAV9, ANOVA.

FIGS. 5A-5B depict representative images of brain sections and their quantitative analysis after intravenous administration of AAV9, AAV.CPP.16 and AAV.CPP.21 in C57BL/6J inbred mice. All candidate AAVs express nuclear red fluorescent protein (RFP) as reporter. The dose of AAV is 1×10^{12} vg per animal. Each white dot in FIG. 5A represents a RFP-labeled cell. In FIG. 5B, * $P < 0.05$, *** $P < 0.001$, ANOVA.

FIGS. 6A-6B depict representative images of brain sections and their quantitative analysis after intravenous administration of AAV9, AAV.CPP.16 and AAV.CPP.21 in BALB/cJ inbred mice. All candidate AAVs express nuclear red fluorescent protein (RFP) as reporter. The dose of AAV is 1×10^{12} vg per animal. Each white dot in FIG. 6A represents a RFP-labeled cell. In FIG. 6B, *** $P < 0.001$, ANOVA.

FIGS. 7A-7B depict representative images of brain sections and their quantitative analysis after intravenous administration of high-dose AAV.CPP.16 and AAV.CPP.21 in C57BL/6J inbred mice. Both candidate AAVs express nuclear red fluorescent protein (RFP) as reporter. The dose of AAV is 4×10^{12} vg per animal. Each white dot in FIG. 7A represents a RFP-labeled cell. In FIG. 7B, * $P < 0.05$, Student test.

FIG. 8A shows AAV.CPP.16 and AAV.CPP.21 transduce adult neurons (labeled by a NeuN antibody) across multiple brain regions in mice including the cortex, midbrain and hippocampus. Transduced neurons are co-labeled by NeuN antibody and RFP. AAVs of 4×10^{12} vg were administered intravenously in adult C57BL/6J mice (6 weeks old).

FIG. 8B depicts that AAV.CPP.16 and AAV.CPP.21 show enhanced ability vs. AAV9 in targeting the spinal cord and motor neurons in mice. AAVs of 4×10^{10} vg were administered intravenously into neonate mice (1 day after birth). Motor neurons in the ventral horn of the spinal cord were visualized using CHAT antibody staining. Co-localization of RFP and CHAT signals suggests specific transduction of the motor neurons.

FIG. 9A depicts that AAV.CPP.16 shows enhanced ability vs. AAV9 in targeting the heart in adult mice. AAVs of 1×10^{11} vg were administered intravenously in adult C57BL/

6J mice (6 weeks old). Percentage of RFP-labeled cells relative to all DAPI-stained cells is presented. * $P < 0.05$, Student test.

FIG. 9B depicts that AAV.CPP.16 shows enhanced ability vs. AAV9 in targeting the skeletal muscle in adult mice. AAVs of 1×10^{11} vg were administered intravenously in adult C57BL/6J mice (6 weeks old). Percentage of RFP-labeled cells relative to all DAPI-stained cells is presented. * $P < 0.05$, Student test.

FIG. 9C depicts that AAV.CPP.16 shows enhanced ability vs. AAV9 in targeting the dorsal root ganglion (DRG) in adult mice. AAVs of 1×10^{11} vg were administered intravenously in adult C57BL/6J mice (6 weeks old). Percentage of RFP-labeled cells relative to all DAPI-stained cells is presented. * $P < 0.05$, Student test.

FIG. 10A depicts that AAV.CPP.16 and AAV.CPP.21 show enhanced ability vs. AAV9 to transduce brain cells in primary visual cortex after intravenous administration in non-human primates. 2×10^{13} vg/kg AAVs-CAG-AADC (as reporter gene) were injected intravenously into 3 months old cynomolgus monkeys with low pre-existing neutralizing antibody. AAV-transduced cells (shown in black) were visualized using antibody staining against AADC. Squared areas in the left panels are enlarged as in the right panels. AAV.CPP.16 transduced significantly more cells vs. AAV9. AAV.CPP.21 also transduced more cell vs. AAV9 although its effect was less evident in comparison with AAV.CPP.16.

FIG. 10B depicts that AAV.CPP.16 and AAV.CPP.21 show enhanced ability vs. AAV9 to transduce brain cells in parietal cortex after intravenous administration in non-human primates. 2×10^{13} vg/kg AAVs-CAG-AADC (as reporter gene) were injected intravenously into 3 months old cynomolgus monkeys with low pre-existing neutralizing antibody. AAV-transduced cells (shown in black) were visualized using antibody staining against AADC. Squared areas in the left panels are enlarged as in the right panels. AAV.CPP.16 transduced significantly more cells vs. AAV9. AAV.CPP.21 also transduced more cell vs. AAV9 although its effect was less evident in comparison with AAV.CPP.16.

FIG. 10C depicts that AAV.CPP.16 and AAV.CPP.21 show enhanced ability vs. AAV9 to transduce brain cells in thalamus after intravenous administration in non-human primates. 2×10^{13} vg/kg AAVs-CAG-AADC (as reporter gene) were injected intravenously into 3 months old cynomolgus monkeys with low pre-existing neutralizing antibody. AAV-transduced cells (shown in black) were visualized using antibody staining against AADC. Squared areas in the left panels are enlarged as in the right panels. AAV.CPP.16 transduced significantly more cells vs. AAV9. AAV.CPP.21 also transduced more cell vs. AAV9 although its effect was less evident in comparison with AAV.CPP.16.

FIG. 10D depicts that AAV.CPP.16 and AAV.CPP.21 show enhanced ability vs. AAV9 to transduce brain cells in cerebellum after intravenous administration in non-human primates. 2×10^{13} vg/kg AAVs-CAG-AADC (as reporter gene) were injected intravenously into 3 months old cynomolgus monkeys with low pre-existing neutralizing antibody. AAV-transduced cells (shown in black) were visualized using antibody staining against AADC. Squared areas in the left panels are enlarged as in the right panels. Both AAV.CPP.16 and AAV.CPP.21 transduced significantly more cells vs. AAV9.

FIGS. 11A-11B depict that AAV.CPP.16 and AAV.CPP.21 do not bind to LY6A. LY6A serves as a receptor for AAV.PHP.B and its variants including AAV.PHP.eB (as in U.S. Pat. No. 9,102,949, US20170166926) and mediates AAV.PHP.eB's robust effect in crossing the BBB in certain

mouse strains (Hordeaux et al. Mol Ther 2019 27(5):912-921; Huang et al. 2019, dx.doi.org/10.1101/538421). Over-expressing mouse LY6A in cultured 293 cells significantly increases binding of AAV.PHP.eB to the cell surface (FIG. 11A). On the contrary, over-expressing LY6A does not increase viral binding for AAV9, AAV.CPP.16 or AAV.CPP.21 (FIG. 11B). This suggests AAV.CPP.16 or AAV.CPP.21 does not share LY6A with AAV.PHP.eB as a receptor.

FIGS. 12A-12C depict that AAV.CPP.21 can be used to systemically deliver a therapeutic gene into brain tumor in a mouse mode of glioblastoma (GBM). As in FIG. 11A, intravenously administered AAV.CPP.21-H2BmCherry was shown to target tumor mass, especially the tumor expanding frontier. In FIG. 11B-11C, using AAV.CPP.21 to systemically deliver the "suicide gene" HSV.TK1 results in shrinkage of brain tumor mass, when combined with the pro-drug ganciclovir. HSV.TK1 turns the otherwise "dormant" ganciclovir into a tumor-killing drug. * $P < 0.05$, Student test.

FIG. 13 depicts that when injected locally into adult mouse brain, AAV.CPP.21 resulted in more widespread and robust transduction of brain tissue in comparison with AAV9. Intracerebral injection of AAVs (1×10^{11} vg) was performed in adult mice (>6 weeks old) and brain tissues were harvested and examined 3 weeks after AAV injection. ** $P < 0.01$, Student test.

DETAILED DESCRIPTION

Difficulties associated with delivery across the BBB have hindered development of therapeutic agents to treat brain disorders including cancer and neurodegenerative disorders. Adeno-associated virus (AAV) has emerged as an important research and clinical tool for delivering therapeutic genes to the brain, spinal cord and the eye; see, e.g., U.S. Pat. Nos. 9,102,949; 9,585,971; and US20170166926. However, existing AAVs including AAV9 have either limited efficiency in crossing the BBB, or only work in some non-primate species.

Through rational design and targeted screening on the basis of known cell-penetrating peptides (CPPs) (see, e.g., Gomez et al., *Bax-inhibiting peptides derived from Ku70 and cell-penetrating pentapeptides*. Biochem. Soc. Trans. 2007; 35(Pt 4):797-801), targeting sequences have been discovered that, when engineered into the capsid of an AAV, improved the efficiency of gene delivery to the brain by up to three orders of magnitude. These methods were used to engineer one such AAV vector that dramatically reduced tumor size in an animal model of glioblastoma.

Targeting Sequences

The present methods identified a number of potential targeting peptides that enhance permeation through the BBB, e.g., when inserted into the capsid of an AAV, e.g., AAV1, AAV2, AAV8, or AAV9, or when conjugated to a biological agent, e.g., an antibody or other large biomolecule, either chemically or via expression as a fusion protein.

In some embodiments, the targeting peptides comprise sequences of at least 5 amino acids. In some embodiments, the amino acid sequence comprises at least 4, e.g., 5, contiguous amino acids of the sequences VPALR (SEQ ID NO:1) and VSALK (SEQ ID NO:2).

In some embodiments, the targeting peptides comprise a sequence of $X_1 X_2 X_3 X_4 X_5$, wherein:

- (i) X_1, X_2, X_3, X_4 are any four non-identical amino acids of V, A, L, I, G, P, S, T, or M; and
- (ii) X_5 is K, R, H, D, or E (SEQ ID NO:73).

7

In some embodiments, the targeting peptides comprise sequences of at least 6 amino acids. In some embodiments, the amino acid sequence comprises at least 4, e.g., 5 or 6 contiguous amino acids of the sequences TVPALR (SEQ ID NO:3), TVSALK (SEQ ID NO:4), TVPMLK (SEQ ID NO:12) and TVPTLK (SEQ ID NO:13).

In some embodiments, the targeting peptides comprise a sequence of $X_1 X_2 X_3 X_4 X_5 X_6$, wherein:

- (i) X_1 is T;
- (ii) X_2, X_3, X_4, X_5 are any four non-identical amino acids of V, A, L, I, G, P, S, T, or M; and
- (iii) X_6 is K, R, H, D, or E (SEQ ID NO:74).

In some embodiments, the targeting peptides comprise a sequence of $X_1 X_2 X_3 X_4 X_5 X_6$, wherein:

- (i) X_1, X_2, X_3, X_4 are any four non-identical amino acids from V, A, L, I, G, P, S, T, or M;
- (ii) X_5 is K, R, H, D, or E; and
- (iii) X_6 is E or D (SEQ ID NO:75).

In some embodiments, the targeting peptides comprise sequences of at least 7 amino acids. In some embodiments, the amino acid sequence comprises at least 4, e.g., 5, 6, or 7 contiguous amino acids of the sequences FTVSALK (SEQ ID NO:5), LTVSALK (SEQ ID NO:6), TVSALFK (SEQ ID NO:8), TVPALFR (SEQ ID NO:9), TVPMLFK (SEQ ID NO:10) and TVPTLFK (SEQ ID NO:11). In some other embodiments, the targeting peptides comprise a sequence of $X_1 X_2 X_3 X_4 X_5 X_6 X_7$, wherein:

- (i) X_1 is F, L, W, or Y;
- (ii) X_2 is T;
- (iii) X_3, X_4, X_5, X_6 are any four non-identical amino acids of V, A, L, I, G, P, S, T, or M; and
- (iv) X_7 is K, R, H, D, or E (SEQ ID NO:76).

In some embodiments, the targeting peptides comprise a sequence of $X_1 X_2 X_3 X_4 X_5 X_6 X_7$, wherein:

- (i) X_1 is T;
- (ii) X_2, X_3, X_4, X_5 are any four non-identical amino acids of V, A, L, I, G, P, S, T, or M;
- (iii) X_6 is K, R, H, D, or E; and
- (iv) X_7 is E or D (SEQ ID NO:77).

In some embodiments, the targeting peptides comprise a sequence of $X_1 X_2 X_3 X_4 X_5 X_6 X_7$, wherein:

- (i) X_1, X_2, X_3, X_4 are any four non-identical amino acids of V, A, L, I, G, P, S, T, or M;
- (ii) X_5 is K, R, H, D, or E;
- (iii) X_6 is E or D; and
- (iv) X_7 is A or I (SEQ ID NO:78).

In some embodiments, the targeting peptides comprise a sequence of V[S/p][A/m/t]/L (SEQ ID NO:79), wherein the upper case letters are preferred at that position. In some embodiments, the targeting peptides comprise a sequence of TV[S/p][A/m/t]/L (SEQ ID NO:80). In some embodiments, the targeting peptides comprise a sequence of TV[S/p][A/m/t]/LK (SEQ ID NO:81). In some embodiments, the targeting peptides comprise a sequence of TV[S/p][A/m/t]/LFK. (SEQ ID NO:82).

In some embodiments, the targeting peptide does not consist of VPALR (SEQ ID NO:1) or VSALK (SEQ ID NO:2).

Specific exemplary amino acid sequences that include the above mentioned 5, 6, or 7-amino acid sequences are listed in Table 1.

8

TABLE 1

Targeting Sequences	
SEQ ID NO:	Targeting Peptide Sequence
1.	VPALR
2.	VSALK
3.	TVPALR
4.	TVSALK
5.	FTVSALK
6.	LTVSALK
7.	TFVSALK
8.	TVSALFK
9.	TVPALFR
10.	TVPMLFK
11.	TVPTLFK
12.	TVPMLK
13.	TVPTLK
14.	VPMLK
15.	VPTLK
16.	VPMLKE
17.	VPTLKD
18.	VPALRD
19.	VSALKE
20.	VSALKD
21.	TAVSLK
22.	TALVSK
23.	TVLSAK
24.	TLVSAK
25.	TMVPLK
26.	TMLVPK
27.	TVLPMK
28.	TLVPMK
29.	TTVPLK
30.	TTLVPK
31.	TVLPTK
32.	TLVPTK
33.	TAVPLR
34.	TALVPR
35.	TVLPAR
36.	TLVPAR
37.	TAVSLKE
38.	TALVSKE
39.	TVLSAKE
40.	TLVSAKE
41.	TMVPLKE
42.	TMLVPKE
43.	TVLPMKE
44.	TLVPMKE
45.	TTVPLKD
46.	TTLVPKD
47.	TVLPTKD
48.	TLVPTKD
49.	TAVPLRD
50.	TALVPRD
51.	TVLPARD
52.	TLVPARD
53.	TAVSLFK
54.	TALVSFK
55.	TVLSAFK
56.	TLVSAFK
57.	TMVPLFK
58.	TMLVPFK
59.	TVLPMFK
60.	TLVPMFK
61.	TTVPLFK
62.	TTLVPFK
63.	TVLPTFK
64.	TLVPTFK
65.	TAVPLFR
66.	TALVPFR
67.	TVLPAFR
68.	TLVPAFR

Targeting peptides including reversed sequences can also be used, e.g., KLASVT (SEQ ID NO:83) and KFLASVT (SEQ ID NO:84).

Targeting peptides disclosed herein can be modified according to the methods known in the art for producing peptidomimetics. See, e.g., Qvit et al., Drug Discov Today.

2017 February; 22(2): 454-462; Farhadi and Hashemian, Drug Des Devel Ther. 2018; 12: 1239-1254; Avan et al., Chem. Soc. Rev., 2014, 43, 3575-3594; Pathak, et al., Indo American Journal of Pharmaceutical Research, 2015. 8; Kazmierski, W. M., ed., Peptidomimetics Protocols, Human Press (Totowa NJ 1998); Goodman et al., eds., Houben-Weyl Methods of Organic Chemistry: Synthesis of Peptides and Peptidomimetics, Thiele Verlag (New York 2003); and Mayo et al., J. Biol. Chem., 278:45746 (2003). In some cases, these modified peptidomimetic versions of the peptides and fragments disclosed herein exhibit enhanced stability in vivo, relative to the non-peptidomimetic peptides.

Methods for creating a peptidomimetic include substituting one or more, e.g., all, of the amino acids in a peptide sequence with D-amino acid enantiomers. Such sequences are referred to herein as "retro" sequences. In another method, the N-terminal to C-terminal order of the amino acid residues is reversed, such that the order of amino acid residues from the N-terminus to the C-terminus of the original peptide becomes the order of amino acid residues from the C-terminus to the N-terminus in the modified peptidomimetic. Such sequences can be referred to as "inverso" sequences.

Peptidomimetics can be both the retro and inversed versions, i.e., the "retro-inverso" version of a peptide disclosed herein. The new peptidomimetics can be composed of D-amino acids arranged so that the order of amino acid residues from the N-terminus to the C-terminus in the peptidomimetic corresponds to the order of amino acid residues from the C-terminus to the N-terminus in the original peptide.

Other methods for making a peptidomimetic include replacing one or more amino acid residues in a peptide with a chemically distinct but recognized functional analog of the amino acid, i.e., an artificial amino acid analog. Artificial amino acid analogs include β -amino acids, β -substituted β -amino acids (" β^3 -amino acids"), phosphorous analogs of amino acids, such as ∇ -amino phosphonic acids and ∇ -amino phosphinic acids, and amino acids having non-peptide linkages. Artificial amino acids can be used to create peptidomimetics, such as peptoid oligomers (e.g., peptoid amide or ester analogues), β -peptides, cyclic peptides, oligoureia or oligocarbamate peptides; or heterocyclic ring molecules. Exemplary retro-inverso targeting peptidomimetics include KLASVT and KFLASVT, wherein the

sequences include all D-amino acids. These sequences can be modified, e.g., by biotinylation of the amino terminus and amidation of the carboxy terminus.

AAVs

Viral vectors for use in the present methods and compositions include recombinant retroviruses, adenovirus, adeno-associated virus, alphavirus, and lentivirus, comprising the targeting peptides described herein and optionally a transgene for expression in a target tissue.

A preferred viral vector system useful for delivery of nucleic acids in the present methods is the adeno-associated virus (AAV). AAV is a tiny non-enveloped virus having a 25 nm capsid. No disease is known or has been shown to be associated with the wild type virus. AAV has a single-stranded DNA (ssDNA) genome. AAV has been shown to exhibit long-term episomal transgene expression, and AAV has demonstrated excellent transgene expression in the brain, particularly in neurons. Vectors containing as little as 300 base pairs of AAV can be packaged and can integrate. Space for exogenous DNA is limited to about 4.7 kb. An AAV vector such as that described in Tratschin et al., Mol. Cell. Biol. 5:3251-3260 (1985) can be used to introduce DNA into cells. A variety of nucleic acids have been introduced into different cell types using AAV vectors (see for example Hermonat et al., Proc. Natl. Acad. Sci. USA 81:6466-6470 (1984); Tratschin et al., Mol. Cell. Biol. 4:2072-2081 (1985); Wondisford et al., Mol. Endocrinol. 2:32-39 (1988); Tratschin et al., J. Virol. 51:611-619 (1984); and Flotte et al., J. Biol. Chem. 268:3781-3790 (1993). There are numerous alternative AAV variants (over 100 have been cloned), and AAV variants have been identified based on desirable characteristics. In some embodiments, the AAV is AAV1, AAV2, AAV4, AAV5, AAV6, AV6.2, AAV7, AAV8, AAV9, rh.10, rh.39, rh.43 or CSP3; for CNS use, in some embodiments the AAV is AAV1, AAV2, AAV4, AAV5, AAV6, AAV8, or AAV9. As one example, AAV9 has been shown to somewhat efficiently cross the blood-brain barrier. Using the present methods, the AAV capsid can be genetically engineered to increase permeation across the BBB, or into a specific tissue, by insertion of a targeting sequence as described herein into the capsid protein, e.g., into the AAV9 capsid protein VP1 between amino acids 588 and 589.

An exemplary wild type AAV9 capsid protein VP1 (Q6JC40-1) sequence is as follows:

(SEQ ID NO: 85)

10	20	30	40	50
MAADGYLPDW	LEDNLSEGR	EWALKPGAP	QPKANQHQD	NARGLVLPGY
60	70	80	90	100
KYLGPGNGLD	KGEPVNAADA	AALEHDKAYD	QQLKAGDNPY	LKYNHADAEEF
110	120	130	140	150
QERLKEDTSF	GGNLGRAVFQ	AKKRLLEPLG	LVVEEAAKTAP	GKKRPVEQSP
160	170	180	190	200
QEPDSSAGIG	KSGAQPAKKR	LNFGQTGDE	SVPDPQPIGE	PPAAPSGVGS
210	220	230	240	250
LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRI	TTSTRTWALP
260	270	280	290	300
TYNNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYFDNFRFH	CHESPRDWQR
310	320	330	340	350
LINNNGWFRP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY
360	370	380	390	400
QLPYVLGSAH	EGCLPPFPAD	VEMIPQGYL	TLNDGSQAVG	RSSFYCLEYF

- continued

410 420 430 440 450
 PSQMLRTGNN PQFSYEFENV PFHSSYAHSQ SLDRLMNPLI DQYLYYLSKT

 460 470 480 490 500
 INSGGQNQQT LKFSVAGPSN MAVQGRNYIP GPSYRQQRVS TTVTQNNNSE

 510 520 530 540 550
 FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS LIFGKQGTGR

 560 570 580 590 600
 DNVDAKVM I TNEEEIKTTN PVATESYGQV ATNHQSAQAQ AQTGWVQNQG

 610 620 630 640 650
 ILPGMVWQDR DVYLQGP IWA KIPHTDGNFH PSPLMGGFGM KHPPPQILIK

 660 670 680 690 700
 NTPVPADPPT AFNKDKLNSF ITQYSTGQVS VEIEWELQKE NSKRWNPEIQ

 710 720 730
 YTSNYYKSNN VEFVNTTEGV YSEPRPIGTR YLTRNL

Thus provided herein are AAV that include one or more of the targeting peptide sequences described herein, e.g., an AAV comprising a capsid protein comprising a targeting sequence described herein, e.g., a capsid protein comprising SEQ ID NO:1 wherein a targeting peptide sequence has been inserted into the sequence, e.g., between amino acids 588 and 589.

In some embodiments, the AAV also includes a transgene sequence (i.e., a heterologous sequence), e.g., a transgene encoding a therapeutic agent, e.g., as described herein or as known in the art, or a reporter protein, e.g., a fluorescent protein, an enzyme that catalyzes a reaction yielding a detectable product, or a cell surface antigen. The transgene is preferably linked to sequences that promote/drive expression of the transgene in the target tissue.

Exemplary transgenes for use as therapeutics include neuronal apoptosis inhibitory protein (NAIP), nerve growth factor (NGF), glial-derived growth factor (GDNF), brain-derived growth factor (BDNF), ciliary neurotrophic factor (CNTF), tyrosine hydroxylase (TH), GTP-cyclohydrolase (GTPCH), amino acid decarboxylase (AADC), aspartoacylase (ASPA), blood factors, such as β -globin, hemoglobin, tissue plasminogen activator, and coagulation factors; colony stimulating factors (CSF); interleukins, such as IL-1, IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, etc.; growth factors, such as keratinocyte growth factor (KGF), stem cell factor (SCF), fibroblast growth factor (FGF, such as basic FGF and acidic FGF), hepatocyte growth factor (HGF), insulin-like growth factors (IGFs), bone morphogenetic protein (BMP), epidermal growth factor (EGF), growth differentiation factor-9 (GDF-9), hepatoma derived growth factor (HDGF), myostatin (GDF-8), nerve growth factor (NGF), neurotrophins, platelet-derived growth factor (PDGF), thrombopoietin (TPO), transforming growth factor alpha (TGF- α), transforming growth factor beta (TGF- β), and the like; soluble receptors, such as soluble TNF- α receptors, soluble VEGF receptors, soluble interleukin receptors (e.g., soluble IL-1 receptors and soluble type II IL-1 receptors), soluble gamma/delta T cell receptors, ligand-binding fragments of a soluble receptor, and the like; enzymes, such as α -glucosidase, imiglucrase, β -glucocerebrosidase, and alglucerase; enzyme activators, such as tissue plasminogen activator; chemokines, such as IP-10, monokine induced by interferon-gamma (Mig), Groa/IL-8, RANTES, MIP-1 α , MIP-10, MCP-1, PF-4, and the like; angiogenic agents, such as vascular endothelial growth factors (VEGFs, e.g., VEGF121, VEGF165, VEGF-C, VEGF-2), transforming

20 growth factor-beta, basic fibroblast growth factor, glioma-derived growth factor, angiogenin, angiogenin-2; and the like; anti-angiogenic agents, such as a soluble VEGF receptor; protein vaccine; neuroactive peptides, such as nerve growth factor (NGF), bradykinin, cholecystokinin, gastrin, secretin, oxytocin, gonadotropin-releasing hormone, beta-endorphin, enkephalin, substance P, somatostatin, prolactin, galanin, growth hormone-releasing hormone, bombesin, dynorphin, warfarin, neurotensin, motilin, thyrotropin, neuropeptide Y, luteinizing hormone, calcitonin, insulin, glucagons, vasopressin, angiotensin II, thyrotropin-releasing hormone, vasoactive intestinal peptide, a sleep peptide, and the like; thrombolytic agents; atrial natriuretic peptide; relaxin; glial fibrillary acidic protein; follicle stimulating hormone (FSH); human alpha-1 antitryp sin; leukemia inhibitory factor (LIF); transforming growth factors (TGFs); tissue factors, luteinizing hormone; macrophage activating factors; tumor necrosis factor (TNF); neutrophil chemotactic factor (NCF); nerve growth factor; tissue inhibitors of metalloproteinases; vasoactive intestinal peptide; angiogenin; angiotropin; fibrin; hirudin; IL-1 receptor antagonists; and the like. Some other examples of protein of interest include ciliary neurotrophic factor (CNTF); neurotrophins 3 and 4/5 (NT-3 and 4/5); glial cell derived neurotrophic factor (GDNF); aromatic amino acid decarboxylase (AADC); hemophilia related clotting proteins, such as Factor VIII, Factor IX, Factor X; dystrophin or mini-dystrophin; lysosomal acid lipase; phenylalanine hydroxylase (PAH); glycogen storage disease-related enzymes, such as glucose-6-phosphatase, acid maltase, glycogen debranching enzyme, muscle glycogen phosphorylase, liver glycogen phosphorylase, muscle phosphofructokinase, phosphorylase kinase (e.g., PHKA2), glucose transporter (e.g., GLUT2), aldolase A, β -enolase, and glycogen synthase; lysosomal enzymes (e.g., beta-N-acetylhexosaminidase A); and any variants thereof.

55 The transgene can also encode an antibody, e.g., an immune checkpoint inhibitory antibody, e.g., to PD-L1, PD-1, CTLA-4 (Cytotoxic T-Lymphocyte-Associated Protein-4; CD152); LAG-3 (Lymphocyte Activation Gene 3; CD223); TIM-3 (T-cell Immunoglobulin domain and Mucin domain 3; HAVCR2); TIGIT (T-cell Immunoreceptor with Ig and ITIM domains); B7-H3 (CD276); VSIR (V-set immunoregulatory receptor, aka VISTA, B7H5, C10orf54); BTLA 30 (B- and T-Lymphocyte Attenuator, CD272); GARP (Glycoprotein A Repetitions; Predominant; PVRIG (PVR related immunoglobulin domain containing); or VTCN1 (Vset domain containing T cell activation inhibitor 1, aka B7-H4).

13

Other transgenes can include small or inhibitory nucleic acids that alter/reduce expression of a target gene, e.g., siRNA, shRNA, miRNA, antisense oligos, or long non-coding RNAs that alter gene expression (see, e.g., WO2012087983 and US20140142160), or CRISPR Cas9/cas12a and guide RNAs.

The virus can also include one or more sequences that promote expression of a transgene, e.g., one or more promoter sequences; enhancer sequences, e.g., 5' untranslated region (UTR) or a 3' UTR; a polyadenylation site; and/or insulator sequences. In some embodiments, the promoter is a brain tissue specific promoter, e.g., a neuron-specific or glia-specific promoter. In certain embodiments, the promoter is a promoter of a gene selected to from: neuronal nuclei (NeuN), glial fibrillary acidic protein (GFAP), MeCP2, adenomatous polyposis *coli* (APC), ionized calcium-binding adapter molecule 1 (Iba-1), synapsin I (SYN), calcium/calmodulin-dependent protein kinase II, tubulin alpha I, neuron-specific enolase and platelet-derived growth factor beta chain. In some embodiments, the promoter is a pan-cell type promoter, e.g., cytomegalovirus (CMV), beta glucuronidase, (GUSB), ubiquitin C (UBC), or rous sarcoma virus (RSV) promoter. The woodchuck hepatitis virus post-transcriptional response element (WPRE) can also be used.

In some embodiments, the AAV also has one or more additional mutations that increase delivery to the target tissue, e.g., the CNS, or that reduce off-tissue targeting, e.g., mutations that decrease liver delivery when CNS, heart, or muscle delivery is intended (e.g., as described in Pulicherla et al. (2011) Mol Ther 19:1070-1078); or the addition of other targeting peptides, e.g., as described in Chen et al. (2008) Nat Med 15:1215-1218 or Xu et al., (2005) Virology 341:203-214 or U.S. Pat. Nos. 9,102,949; 9,585,971; and US20170166926. See also Gray and Samulski (2011) "Vector design and considerations for CNS applications," in Gene Vector Design and Application to Treat Nervous System Disorders ed. Glorioso J., editor. (Washington, DC: Society for Neuroscience;) 1-9, available at sfn.org/~media/SfN/Documents/Short%20Courses/20110%20Short%20Course%20I/2_011_SC1_Gray.ashx.

Targeting Peptides as Tags/Fusions

The targeting peptides described herein can also be used to increase permeation of other (heterologous) molecules across the BBB, e.g., by conjugation to the molecule, or by expression as part of a fusion protein, e.g., with an antibody or other large biomolecule. These can include genome editing proteins or complexes (e.g., TALEs, ZFNs, Base editors, and CRISPR RNPs comprising a gene editing protein such as Cas9 or Cas12a, fused to a peptide described herein (e.g., at the N terminus, C terminus, or internally) and a guide RNA), in addition to therapeutic agents or reporters as described herein as well as those listed in Table 2. The fusions/complexes do not comprise any other sequences from Ku70, e.g., comprise heterologous non-Ku70 sequences, and are not present in nature.

In some embodiments, targeting sequences used as part of a non-AAV fusion protein do not comprise or consist of VPALR (SEQ ID NO:1) or VSALK (SEQ ID NO:2).

Methods of Use

The methods and compositions described herein can be used to deliver any composition, e.g., a sequence of interest to a tissue, e.g., to the central nervous system (brain), heart, muscle, or dorsal root ganglion or spinal cord (peripheral nervous system). In some embodiments, the methods include delivery to specific brain regions, e.g., cortex, cerebellum, hippocampus, substantia nigra, amygdala. In some

14

embodiments, the methods include delivery to neurons, astrocytes, glial cells, or cardiomyocytes.

In some embodiments, the methods and compositions, e.g., AAVs, are used to deliver a nucleic acid sequence to a subject who has a disease, e.g., a disease of the CNS; see, e.g., U.S. Pat. Nos. 9,102,949; 9,585,971; and US20170166926. In some embodiments, the subject has a condition listed in Table 2; in some embodiments, the vectors are used to deliver a therapeutic agent listed in Table 2 for treating the corresponding disease listed in Table 2. The therapeutic agent can be delivered as a nucleic acid, e.g. via a viral vector, wherein the nucleic acid encodes a therapeutic protein or other nucleic acid such as an antisense oligo, siRNA, shRNA, and so on; or as a fusion protein/complex with a targeting peptide as described herein.

TABLE 2

Diseases		
Examples of diseases	Tissue targeted	Therapeutic agent
Parkinson's disease	CNS	GDNF, AADC
Alzheimer's disease	CNS	Tau antibody, APP antibody
Huntington's disease	CNS	miRNA targeting HTT
Amyotrophic lateral sclerosis	CNS	shRNA targeting SOD
Multiple sclerosis	CNS	IFN-beta
Epilepsy	CNS	Neuropeptide Y
Stroke	CNS	IGF-1, osteopontin
Brain cancer	CNS	HSV.TK1, PD-1/PD-L1 antibody
Spinocerebellar ataxia	CNS	RNAi targeting ataxin
Canavan disease	CNS	ASPA
Metachromatic leukodystrophy	Nervous systems	ARSA, PSAP
Spinal muscular atrophy	Neuromuscular system	SMN1
Friedreich's ataxia	Nervous systems, heart	Frataxin
X-linked myotubular myopathy	Neuromuscular system	MTM1
Pompe disease	Lysosome (global including CNS)	GAA
Barth syndrome	Heart, muscle	TAZ
Duchenne muscular dystrophy	Muscle	dystrophin
Wilson's disease	Brain, liver	ATP7B
Crigler-Najjar syndrome type 1	Liver	UGT1A1

In some embodiments, the compositions and methods are used to treat brain cancer. Brain cancers include gliomas (e.g., glioblastoma multiforme (GBM)), metastases (e.g., from lung, breast, melanoma, or colon cancer), meningiomas, pituitary adenomas, and acoustic neuromas. The compositions include a targeting peptide linked to an anticancer agent, e.g., a "suicide gene" that induces apoptosis in a target cell (e.g., HSV.TK1, Cytosine Deaminase (CD) from Herpes simplex virus or *Escherichia coli*, or *Escherichia coli* purine nucleoside phosphorylase (PNP)/fludarabine; see Krohne et al., Hepatology. 2001 September; 34(3):511-8; Dey and Evans, "Suicide Gene Therapy by Herpes Simplex Virus-1 Thymidine Kinase (HSV-TK)" (2011) DOI: 10.5772/18544) an immune checkpoint inhibitory antibody as known in the art or described herein. For example, an AAV vector comprising a targeting peptide as described herein can be used to deliver the "suicide gene" HSV.TK1 to a brain tumor. HSV.TK1 turns the otherwise "dormant" ganciclovir into a tumor-killing drug. Thus the methods can include systemically, e.g., intravenously, administering an AAV (e.g.,

AAV9) comprising a targeting peptide as described herein and encoding HSV.TK1, and the pro-drug ganciclovir, to a subject who has been diagnosed with brain cancer. Pharmaceutical Compositions and Methods of Administration

The methods described herein include the use of pharmaceutical compositions comprising the targeting peptides as an active ingredient.

Pharmaceutical compositions typically include a pharmaceutically acceptable carrier. As used herein the language "pharmaceutically acceptable carrier" includes saline, solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like, compatible with pharmaceutical administration.

Pharmaceutical compositions are typically formulated to be compatible with its intended route of administration. Examples of routes of administration include parenteral, e.g., intravenous, intraarterial, subcutaneous, intraperitoneal intramuscular or injection or infusion administration. Delivery can thus be systemic or localized.

Methods of formulating suitable pharmaceutical compositions are known in the art, see, e.g., *Remington: The Science and Practice of Pharmacy*, 21st ed., 2005; and the books in the series *Drugs and the Pharmaceutical Sciences: a Series of Textbooks and Monographs* (Dekker, NY). For example, solutions or suspensions used for parenteral application can include the following components: a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerine, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl parabens; antioxidants such as ascorbic acid or sodium bisulfite; chelating agents such as ethylenediaminetetraacetic acid; buffers such as acetates, citrates or phosphates and agents for the adjustment of tonicity such as sodium chloride or dextrose. pH can be adjusted with acids or bases, such as hydrochloric acid or sodium hydroxide. The parenteral preparation can be enclosed in ampoules, disposable syringes or multiple dose vials made of glass or plastic.

Pharmaceutical compositions suitable for injectable use can include sterile aqueous solutions (where water soluble) or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersion. For intravenous administration, suitable carriers include physiological saline, bacteriostatic water, Cremophor EL™ (BASF, Parsippany, NJ) or phosphate buffered saline (PBS). In all cases, the composition must be sterile and should be fluid to the extent that easy syringability exists. It should be stable under the conditions of manufacture and storage and must be preserved against the contaminating action of microorganisms such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), and suitable mixtures thereof. The proper fluidity can be maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. Prevention of the action of microorganisms can be achieved by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, ascorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars, polyalcohols such as mannitol, sorbitol,

sodium chloride in the composition. Prolonged absorption of the injectable compositions can be brought about by including in the composition an agent that delays absorption, for example, aluminum monostearate and gelatin.

Sterile injectable solutions can be prepared by incorporating the active compound in the required amount in an appropriate solvent with one or a combination of ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the active compound into a sterile vehicle, which contains a basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum drying and freeze-drying, which yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof.

In one embodiment, the therapeutic compounds are prepared with carriers that will protect the therapeutic compounds against rapid elimination from the body, such as a controlled release formulation, including implants and microencapsulated delivery systems. Biodegradable, biocompatible polymers can be used, such as ethylene vinyl acetate, polyanhydrides, polyglycolic acid, collagen, polyorthoesters, and polylactic acid. Such formulations can be prepared using standard techniques, or obtained commercially, e.g., from Alza Corporation and Nova Pharmaceuticals, Inc. Liposomal suspensions (including liposomes targeted to selected cells with monoclonal antibodies to cellular antigens) can also be used as pharmaceutically acceptable carriers. These can be prepared according to methods known to those skilled in the art, for example, as described in U.S. Pat. No. 4,522,811.

The pharmaceutical compositions can be included in a kit, container, pack, or dispenser together with instructions for administration. For example, the compositions comprising an AAV comprising a targeting peptide as described herein and a nucleic acid encoding HSV.TK1 can be provided in a kit with ganciclovir.

EXAMPLES

The invention is further described in the following examples, which do not limit the scope of the invention described in the claims.

Materials and Methods

The following materials and methods were used in the Examples below.

1. Generation of Capsid Variants

To generate the capsid variant plasmids, DNA fragments that encode the cell-penetrating peptides (Table 3) were synthesized (GenScript), and inserted into the backbone of the AAV9 Rep-cap plasmid (pRC9) between amino acid position 588 and 589 (VP1 amino acid numbering), using CloneEZ seamless cloning technology (GenScript). CPPs BIP1 (VPALR, SEQ ID NO:1) and BIP2 (VSALK, SEQ ID NO:2), as well as their derivatives such as TVSALK (SEQ ID NO:4) in AAV.CPP.16 and TVSALFK (SEQ ID NO:8) in AAV.CPP.21, are derived from the Ku70 proteins, of which the sequences are provided as below:

```
Human Ku70 MSGWESYYKTEGDEEAEEEEQEEENLEASGDYKYSGRDSLIFLVDASKAMFESQSEDELTPF 60
Mouse Ku70 MSEWESYYKTEGEEEEEE--EESPD TGGEYKYSGRDSLIFLVDASRAMFESQGEDELTPF 58
Rat Ku70 MSEWESYYKTEGEEEEEE--EQSPD TNGEYKYSGRDSLIFLVDASRAMFESQGEDELTPF 58
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-continued

Human Ku70 DMSIQCIQSVYISKIISDRDLLAVVFYGTEDKNSVNFKNYVLQELDNPAGAKRILELD 120

Mouse Ku70 DMSIQCIQSVYTSKIISDRDLLAVVFYGTEDKNSVNFKNYVLQDLNPGAKRVLELD 118

Rat Ku70 DMSIQCIQSVYTSKIISDRDLLAVVFYGTEDKNSVNFKSIYVLQDLNPGAKRVLELD 118

Human Ku70 QFKGQQGQKRFQDMMGHGSDYSLSEVLWVCANLFSQVQFKMSHKRIMLFTNEDNPHGNDS 180

Mouse Ku70 QFKGQQGQKHFRTDVGHSYSLSEVLWVCANLFSQVQFKMSHKRIMLFTNEDDPHGRDS 178

Rat Ku70 RFKGQQGQKHFRTDIGHGSDYSLSEVLWVCANLFSQVQFKMSHKRIMLFTNEDDPHGNDS 178

Human Ku70 AKASRARTKAGDLRDTGIFLDMHLKKPGGFDISLFYRDIISIAEDEDLRVHFEESKLE 240

Mouse Ku70 AKASRARTKASDLRDTGIFLDMHLKKPGGFDVSVFYRDIITAEDEDLGVHFEESKLE 238

Rat Ku70 AKASRARTKASDLRDTGIFLDMHLKKPGGFDVSLFYRDIISIAEDEDLGVHFEESKLE 238

Human Ku70 DLLRKVRKETRKRALSRLKLNKDIVISVGIYNLVQKALPPPIKLYRETNEPVKTKT 300

Mouse Ku70 DLLRKVRKETKKRVLSRLKFKLGEDVVLVGIYNLVQKANKPPVRLYRETNEPVKTKT 298

Rat Ku70 DLLRKVRKETKKRVLSRLKFKLGKDVALMVGVYNLVQKANKPPVRLYRETNEPVKTKT 298

Human Ku70 RTFNTSTGGLLLPSDTKRSQIYGSQIILEKEETEELKRFDDPGLMLMGFKPLVLLKKHH 360

Mouse Ku70 RTFNVNTGSLLLPSDTKRSQIYGSQIILEKEETEELKRFDEPGLILMGFKPTVMLKKQH 358

Rat Ku70 RTFNVNTGSLLLPSDTKRSQIYGSQIILEKEETEELKRFDEPGLILMGFKPMVMLKNHH 358

Human Ku70 YLRPSLFVYPEESLVIGSSTLFSALLIKCLEKEVAALCRYTPRRNIPPYFVALVPQEEEL 420

Mouse Ku70 YLRPSLFVYPEESLVIGSSTLFSALLTKCVEKEVIAVCRYTPRKNVSPYFVALVPQEEEL 418

Rat Ku70 YLRPSLFVYPEESLVIGSSTLFSALLTKCVEKEVIAVCRYTARKNVSPYFVALVPQEEEL 418

Human Ku70 DDQKIQVTPPGFQLVFLPYADDKRKMPFTEKIMATPEQVGKMKAIVEKLRFTYRSDSFEN 480

Mouse Ku70 DDQNIQVTPPGFQLVFLPYADDKRKVPFTEKVTANQEIDKMKAIQKLRFTYRSDSFEN 478

Rat Ku70 DDQNIQVTPAGFQLVFLPYADDKRKVPFTEKVMANPEQIDKMKAIQKLRFTYRSDSFEN 478

Human Ku70 PVLQQHFRNLEALALDMEPEQAVDLTLPKVEAMNKRGLSLVDEFKELVYPPDYNPEGKV 540

Mouse Ku70 PVLQQHFRNLEALALDMESEQVVDLTLPKVEAIKKRGLSLADEFKELVYPPGYNPEGKV 538

Rat Ku70 PVLQQHFRNLEALALDMESEQVVDLTLPKVEAIKKRGLSLADEFKELVYPPGYNPEGKI 538

Human Ku70 TKRKHDNEGSGSRPKVEYSEELKTHISKGTGKFTVPMLKEACRAYGLKSGLKKQELL 600

Mouse Ku70 AKRKQDDEGSTSKPKVELSEELKAHFRKGTGKLTVPPTLKDICKAHGLKSGPKKQELL 598

Rat Ku70 AKRKADNEGSAKKPKVELSEELKDLFAKGTGKLTVPALRDICKAYGLKSGPKKQELL 598

Human Ku70 EALTKHFQD- 609 (SEQ ID NO: 86)

Mouse Ku70 DALIRHLEKN 608 (SEQ ID NO: 87)

Rat Ku70 EALSRHLEKN 608 (SEQ ID NO: 88)

In addition, VP1 protein sequences for AAV9, AAV.CPP.16 and AAV.CPP.21 are provided as below:

AAV9 MAADGYLPDWLEDNLSEGIREWALKPGAPQPKANQQHQDNARGVLVPGYKYLGPNGGLD 60

AAV.CPP.16 MAADGYLPDWLEDNLSEGIREWALKPGAPQPKANQQHQDNARGVLVPGYKYLGPNGGLD 60

AAV.CPP.21 MAADGYLPDWLEDNLSEGIREWALKPGAPQPKANQQHQDNARGVLVPGYKYLGPNGGLD 60

AAV9 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADAEFQERLKEDTSFGGNLGRAVEQ 120

AAV.CPP.16 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADAEFQERLKEDTSFGGNLGRAVFQ 120

AAV.CPP.21 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADAEFQERLKEDTSFGGNLGRAVEQ 120

AAV9 AKKRLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIGKSGAQPAAKRLNFGQTGDTE 180

-continued

AAV.CPP16 AKKRLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIGKSGAQPAAKRLNFGQTGDTE 180

AAV.CPP21 AKKRLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIGKSGAQPAAKRLNFGQTGDTE 180

AAV9 SVDPDPQPIGEPPAAPSGVGSLTMASGGGAPVADNNEGADGVGSSGNWHCDSQWLGDRVI 240

AAV.CPP16 SVDPDPQPIGEPPAAPSGVGSLTMASGGGAPVADNNEGADGVGSSGNWHCDSQWLGDRVI 240

AAV.CPP21 SVDPDPQPIGEPPAAPSGVGSLTMASGGGAPVADNNEGADGVGSSGNWHCDSQWLGDRVI 240

AAV9 TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTPWGYFDNRFHCHFSRWDQR 300

AAV.CPP16 TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTPWGYFDNRFHCHFSRWDQR 300

AAV.CPP21 TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTPWGYFDNRFHCHFSRWDQR 300

AAV9 LINNNWGFRPKRLNFKLFNIQVKEVTDNNGVKTIANNLSTVQVFTDSYQLPYVLGSAH 360

AAV.CPP16 LINNNWGFRPKRLNFKLFNIQVKEVTDNNGVKTIANNLSTVQVFTDSYQLPYVLGSAH 360

AAV.CPP21 LINNNWGFRPKRLNFKLFNIQVKEVTDNNGVKTIANNLSTVQVFTDSYQLPYVLGSAH 360

AAV9 EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYFPSQMLRTGNMFQFSYEFENV 420

AAV.CPP16 EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYFPSQMLRTGNMFQFSYEFENV 420

AAV.CPP21 EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYFPSQMLRTGNMFQFSYEFENV 420

AAV9 PFHSSYAHQSGLDRLMNPLIDQYLYLSKTINGSGQNQQTLKFSVAGPSNMAVQGRNYIP 480

AAV.CPP16 PFHSSYAHQSGLDRLMNPLIDQYLYLSKTINGSGQNQQTLKFSVAGPSNMAVQGRNYIP 480

AAV.CPP21 PFHSSYAHQSGLDRLMNPLIDQYLYLSKTINGSGQNQQTLKFSVAGPSNMAVQGRNYIP 480

AAV9 GPSYRQQRVSTTQTQNNSEFAWPGASSWALNGRNSLMNPGPAMASHKEGEDRFPPLSGS 540

AAV.CPP16 GPSYRQQRVSTTQTQNNSEFAWPGASSWALNGRNSLMNPGPAMASHKEGEDRFPPLSGS 540

AAV.CPP21 GPSYRQQRVSTTQTQNNSEFAWPGASSWALNGRNSLMNPGPAMASHKEGEDRFPPLSGS 540

AAV9 LIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGGVATNHQSAQ-----AQAQT 593

AAV.CPP16 LIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGGVATNHQSAQTVSAL-KAQAQT 599

AAV.CPP21 LIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGGVATNHQSAQTVSALFKAQAQT 600

AAV9 GWVQNQGILPGMVWQDRDVYLQGPWAKIPHTDGNFHPSPLMGGFGMKHPPQILIKNT 653

AAV.CPP16 GWVQNQGILPGMVWQDRDVYLQGPWAKIPHTDGNFHPSPLMGGFGMKHPPQILIKNT 659

AAV.CPP21 GWVQNQGILPGMVWQDRDVYLQGPWAKIPHTDGNFHPSPLMGGFGMKHPPQILIKNT 660

AAV9 VPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNVEF 713

AAV.CPP16 VPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNVEF 719

AAV.CPP21 VPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNVEF 720

AAV9 AVNTEGVYSEPRPIGTRYLTRNL 736 (SEQ ID NO: 85)

AAV.CPP16 AVNTEGVYSEPRPIGTRYLTRNL 742 (SEQ ID NO: 89)

AAV.CPP21 AVNTEGVYSEPRPIGTRYLTRNL 743 (SEQ ID NO: 90)

2. Recombinant AAV Production

Recombinant AAVs were packaged using standard three-plasmid co-transfection protocol (pRC plasmid, pHelper plasmid and pAAV plasmid). pRC9 (or its variant), pHelper and pAAV carrying a transgene (e.g. nucleus-directed RFP H2B-mCherry driven by an ubiquitous EF1a promoter) were co-transfected into HEK 293T cells using polyethylenimine (PEI, Polysciences). rAAVs vectors were collected from serum-free medium 72 h and 120 h post transfection and from cell at 120 h post transfection. AAV particles in the medium were concentrated using a PEG-precipitation method with 8% PEG-8000 (wt/vol). Cell pellets containing viral particles were resuspended and lysed through sonication. Combined viral vectors from PEG-precipitation and

cell lysates were treated with DNase and RNase at 37° C. for 30 mins and then purified by iodixanol gradient (15%, 25%, 40% and 60%) with ultracentrifugation (VTi 50 rotor, 40,000 r.p.m, 18° C., 1 h). rAAVs were then concentrated using Millipore Amicon filter unit (UFC910008, 100K MWCO) and formulated in Dulbecco's phosphate buffered saline (PBS) containing 0.001% Pluronic F68 (Gibco).

3. AAV Titering

Virus titer was determined by measuring DNase-resistant genome copies using quantitative PCR. pAAV-CAG-GFP was digested with PVUII (NEB) to generate free ends for the plasmid ITRs, and was used for generating a standard curve. Virus samples were incubated with DNase I to eliminate contaminating DNA, followed by sodium hydroxide treat-

21

ment to dissolve the viral capsid and to release the viral genome. Quantitative PCR was performed using an ITR Forward primer 5'-GGAACCCCTAGTGATGGAGTT (SEQ ID NO:91) and an ITR Reverse primer 5'-CGGCCTCAGTGAGCGA (SEQ ID NO:92). Vector titers were normalized to the rAAV-2 reference standard materials (RSMs, ATCC, cat No:VR-1616, Manassas, VA).

4. Administration of AAV in Mice

For intravenous administration, AAV diluted in sterile saline (0.2 ml) was administered through tail vein injection in adult mice (over 6 weeks of age). Animals then survived for three weeks before being euthanized for tissue harvesting. For intracerebral injection, AAV diluted in PBS (10 μ l) was injected using a Hamilton syringe with coordinates from bregma: 1.0 mm right, 0.3 backward, 2.6 mm deep. All animal studies were performed in an AAALAC-accredited facility with IACUC approval.

5. Mouse Tissue Processing

Anesthetized animals were transcardially perfused with cold phosphate buffered saline (PBS) followed by 4% paraformaldehyde (PFA). Tissues were post-fixed in 4% PFA overnight, and then immersed in 30% sucrose solutions for two days prior to embedding and snap-freezing in OCT. Typically, 80 μ m thick brain sections were cut for imaging of native fluorescence, 40 μ m thick brain sections for IHC.

6. In Vitro Human BBB Spheroid Model

Hot 1% agarose (w/v, 50 μ l) was added in a 96-well plate to cool/solidify. Primary human astrocytes (Lonza Bioscience), human brain microvascular pericytes (HBVP, ScienCell Research Laboratories) and human cerebral microvascular endothelial cells (hCMEC/D3; Cedarlane) were then seeded onto the agarose gel in a 1:1:1 ratio (1500 cells of each type). Cells were cultured at 37° C. in a 5% CO₂ incubator for 48-72 hours to allow for spontaneous assembly of multicellular BBB spheroids. A multicellular barrier was reported to form at the periphery of the spheroid, mimicking the BBB. AAVs-H2B-mCherry were added to the culture medium, and 4 days later all spheroids were fixed using 4% PFA, transferred into a Nunc Lab-Tek II thin-glass 8-well chambered coverglass (Thermo Scientific), and imaged using a Zeiss LSM710 confocal microscope. The intensity of RFP signal inside the spheroids was examined and used as a "read-out".

7. AAV Administration in Non-Human Primate (NHP)

All NHP studies were performed by a CRO in an AAALAC-accredited facility with IACUC approval. Cynomolgus monkeys were pre-screened for little or no pre-existing neutralizing antibody against AAV9 (titer of <1:5). AAV diluted in PBS/0.001% F68 was injected intravenously (via cephalic vein or femoral vein) using a peristaltic pump. 3 weeks later, animals were subject to transcardial perfusion with PBS, followed by 4% PFA. Tissues were then collected and processed for paraffin embedding and sectioning.

8. Immunohistochemistry

Floating staining was performed for mouse tissue sections with primary antibodies diluted in PBS containing 10% donkey serum and 2% Triton X-100. Primary antibodies

22

used include: chicken anti-GFP (1:1000); rabbit anti-RFP (1:1000); mouse anti-NeuN (1:500); rat anti-GFAP(1:500); Goat anti-GFAP(1:500); mouse anti-CD31(1:500). Secondary antibodies conjugated to fluorophores of Alexa Fluor 488, Alexa Fluor 555 or Alexa Fluor 647 were applied against the primary antibody's host species at a dilution of 1:200.

For paraffin sections of NHP tissue, DAB staining was performed to visualize cells transduced by AAV-AADC. Rabbit anti-AADC antibody (1:500, Millipore) was used as primary antibody.

9. AAV Binding Assay

HEK293T cells were cultured at 37° C. in a 5% CO₂ incubator. One day after seeding of HEK293T cells in a 24-well plate at a density of 250,000 cells per well, a cDNA plasmid of LY6A was transiently transfected into the cells using a transfection mixture of 200 μ l DMEM (31053028; Gibco), 1 μ g DNA plasmid and 3 μ g of PEI. 48 hours post transfection, cells were placed on ice to chill down for 10 mins. The medium was then changed with 500 μ l ice-cold serum-free DMEM medium containing rAAVs-mCherry at MOI of 10000. After incubating on ice for one hour, cells with presumably AAVs bound to their surface were washed with cold PBS for three times and were then subject to genomic DNA isolation. Cell-binding viral particles were quantified by using qPCR with primers specific to mCherry and normalized to HEK293T genomes using human GCG as reference.

10. Mouse Model of Glioblastoma

All experiments were performed in compliance with protocols approved by the Animal Care and Use Committees (IACUC) at the Brigham and Women's Hospital and Harvard Medical School. Syngeneic immuno-competent C57BL/6 female mice weighing 20+/-1 g (Envigo) were used. GL261-Luc (100,000 mouse glioblastoma cells) resuspended in 2 μ l phosphate buffered saline (PBS) was injected intracranially using 10 μ l syringe with a 26-gauge needle (80075; Hamilton). A stereotactic frame was used to locate the implantation site (coordinates from bregma in mm: 2 right, 0.5 forward, at a depth of 3.5 into cortex). 7 days later, 200 μ l AAV-HSV-TK1 (1E+12 viral genomes, IV) was administered once and ganciclovir (50 mg/kg) was administered daily for 10 days.

Example 1. Modification of AAV9 Capsid

To identify peptide sequences that would enhance permeation of a biomolecule or virus across the blood brain barrier an AAV peptide display technique was used. Individual cell-penetrating peptides, as listed in Table 3, were inserted into the AAV9 capsid between amino acids 588 and 589 (VP1 numbering) as illustrated in FIG. 1A. The insertion was carried out by modifying the RC plasmid, one of the three plasmids co-transfected for AAV packaging; FIG. 1B shows an exemplary schematic of the experiments. Individual AAV variants were produced and screened separately. See Materials and Methods #1-3 for more details.

TABLE 3

	AAV	Name of CPP insert	Amino acid sequence of CPP	#	No. of CPP residues	Viral titer
Initial screening	AAV9	N/A	N/A		N/A	Normal
	AAV.CPP.1	SynB1	RGGRLSYSRRRFSTSTGR	93	18	Low
	AAV.CPP.2	L-2	HARIKPTFRRLKWKY KGKFW	94	20	Low

TABLE 3-continued

	AAV	Name of CPP insert	Amino acid sequence of CPP	#	No. of CPP residues	Viral titer
	AAV.CPP.3	PreS2-TLM	PLSSIFSRIGDP	95	12	Low
	AAV.CPP.4	Transportan 10	AGYLLGKINLKALAA LAKKIL	96	21	Low
	AAV.CPP.5	SAP	VRLPPPVRLLPPPVRLLPPP	97	18	Normal
	AAV.CPP.6	SAP (E)	VELPPPVELPPPVELPPP	98	18	Normal
	AAV.CPP.7	SVM3	KGTYYKKLMRIPLKGT	99	16	Low
	AAV.CPP.8	(PPR)3	PPRPPRPPR	100	9	Normal
	AAV.CPP.9	(PPR)5	PPRPPRPPRPPRPPR	101	15	Low
	AAV.CPP.10	Polyarginine	RRRRRRRR	102	8	Low
	AAV.CPP.11	Bip1	VPALR	1	5	Normal
	AAV.CPP.12	Bip2	VSALK	2	5	Normal
	AAV.CPP.13	DPV15	LRRERQSRLLRRERQSR	103	16	NA
	AAV.CPP.14	HIV-1 Tat	RKKRRQRRR	104	9	NA
Follow-up screening	AAV.CPP.15	Bip1.1	TVPALR (Rat)	3	6	Normal
	AAV.CPP.16	Bip2.1	TVSALK (Syn)	4	6	Normal
	AAV.CPP.17	Bip2.2	FTVSALK (Syn)	5	7	Normal
	AAV.CPP.18	Bip2.3	LTVSALK (Syn)	6	7	Normal
	AAV.CPP.19	Bip2.4	KFTVSALK (Syn)	72	8	Normal
	AAV.CPP.20	Bip2.5	TFVSALK (Syn)	7	7	Normal
	AAV.CPP.21	Bip2.6	TVSALFK (Syn)	8	7	Normal
	AAV.CPP.22	Bip2.6Rat	TVPALFR (Rat)	9	7	Normal

Example 2. First Round of In Vivo Screening

AAVs expressing nuclear RFP (H2B-RFP) were injected intravenously in adult mice with mixed C57B3L/6 and BALB/c genetic background. 3 weeks later, brain tissues were harvested and sectioned to reveal RFP-labelled cells (white dots in FIGS. 2A and 2C, quantified in FIGS. 2B and 2D, respectively). CPPs BIP1 and BTP2 were inserted into the capsids of AAV.CPP.11 and AAV.CPP.12, respectively. See Materials and Methods #4-5 for more details.

Example 3. Optimization of Modified AAV9 Capsids

AAV.CPP.11 and AAV.CPP.12 were further engineered by optimizing the BIP targeting sequences. BIP inserts were derived from the protein Ku70 (See FIG. 3A and Material/Methods #1 for full sequence). The BIP sequence VSALK, which is of “synthetic” origin, was chosen as a study focus to minimize potential species specificity of engineered AAV vectors. AAVs were produced and tested separately for brain transduction efficiency as compared with AAV9 (see FIGS. 3B-C). Percentages of cell transduction in the mouse liver 3 weeks after IV injection of some AAV variants delivering the reporter gene RFP are shown in FIG. 3D. See Materials and Methods #1-5 for more details.

Example 4. In Vitro Model—BBB Permeation Screening

Some of the AAV variants were screened for the ability to cross the human BBB using an in vitro spheroid BBB model. The spheroid contains human microvascular endothelial cells, which form a barrier at the surface, and human pericytes and astrocytes. AAVs carrying nuclear RFP as reporter were assessed for their ability to penetrate from the surrounding medium into the inside of the spheroid and to transduce the cells inside. FIG. 4A shows an experimental schematic. FIGS. 4B-D show results for wt AAV9, AAV.CPP.16, and AAV.CPP.21, respectively, those and other peptides are quantified in FIG. 4E. In this model, peptides

11, 15, 16, and 21 produced the greatest permeation into the spheroids. See Materials and Methods #6 for more details.

Example 5. In Vivo BBB Permeation Screening

AAV.CPP.16 and AAV.CPP.21 were selected for further evaluation in an in vivo model, in experiments performed as described above for Example 2. All AAVs carried nuclear RFP as reporter. Both showed enhanced ability vs. AAV9 to transduce brain cells after intravenous administration in C57BL/6J adult mice (white dots in brain sections in FIG. 5A, quantified in FIG. 5B) and in BALB/c adult mice (white dots in brain sections in FIG. 6A, quantified in FIG. 6B).

High doses of AAV.CPP.16 and AAV.CPP.21 (4×10^{12} vg per mouse, administered IV) resulted in widespread brain transduction in mice. Both AAVs carried nuclear RFP as reporter (white dots in brain sections in FIG. 7A, quantified in FIG. 7B).

Example 6. In Vivo Distribution of Modified AAVs

As shown in FIG. 8A, AAV.CPP.16 and AAV.CPP.21 preferentially targeted neurons (labeled by a NeuN antibody) across multiple brain regions in mice including the cortex, midbrain and hippocampus. Both AAVs carried nuclear RFP as a reporter.

AAV.CPP.16 and AAV.CPP.21 also showed enhanced ability vs. AAV9 in targeting the spinal cord and motor neurons in mice. All AAVs carry nuclear RFP as reporter and were administered intravenously into neonate mice (4×10^{10} vg). Motor neurons were visualized using CHAT antibody staining. Co-localization of RFP and CHAT signals in FIG. 8B suggested specific transduction of the motor neurons.

The relative abilities of AAV-CAG-H2B-RFP and AAV.CPP.16-CAG-H2B-RFP to transduce various tissues in mice was also evaluated. 1×10^{11} vg was injected intravenously. The number of cells transduced was normalized to the number of total cells labeled by DAPI nuclear staining. The results showed that AAV.CPP.16 was more efficient than AAV9 in targeting heart (FIG. 9A); skeletal muscle (FIG. 9B), and dorsal root ganglion (FIG. 9C) tissue in mice.

25

Example 7. BBB Permeation in a Non-Human
Primate Model

2×10¹³ vg/kg AAVs-CAG-AADC (as reporter gene) were injected intravenously into 3-month-old cynomolgus monkeys. AAV-transduced cells (shown in black) were visualized using antibody staining against AADC. As shown in FIGS. 10A-D, AAV.CPP.16 and AAV.CPP.21 showed enhanced ability vs. AAV9 to transduce brain cells after intravenous administration in non-human primates. AAV.CPP.16 transduced significantly more cells than wt AAV9 in the primary visual cortex (FIG. 10A), parietal cortex (FIG. 10B), thalamus (FIG. 10C), and cerebellum (FIG. 10D). See Materials and Methods #7-8 for more details.

Example 8. AAV.CPP.16 and AAV.CPP.21 do not
Bind to LY6A

LY6A serves as a receptor for AAV.PHP.eB and mediates AAV.PHP.eB's robust effect in crossing the BBB in certain mouse strains. Over-expressing mouse LY6A in cultured 293 cells significantly increased binding of AAV.PHP.eB to the cell surface (see FIG. 11A). On the contrary, over-expressing LY6A does not increase viral binding for AAV9, AAV.CPP.16 or AAV.CPP.21 (see FIG. 11B). This suggests AAV.CPP.16 or AAV.CPP.21 does not share LY6A with AAV.PHP.eB as a receptor. See Materials and Methods #9 for more details.

Example 9. Delivering Therapeutic Proteins to the
Brain Using AAV.CPP.21

AAV.CPP.21 was used to systemically deliver the "suicide gene" HSV.TK1 in a mouse model of brain tumor. HSV.TK1

26

turns the otherwise "dormant" ganciclovir into a tumor-killing drug. Intravenously administered AAV.CPP.21-H2BmCherry (FIG. 12A, bottom left and middle right panel) was shown to target tumor mass, especially the tumor expanding frontier. As shown in FIGS. 12B-C, using AAV.CPP.21 to systemically deliver the "suicide gene" HSV.TK1 resulted in shrinkage of brain tumor mass, when combined with the pro-drug ganciclovir. These results show that AAV.CPP.21 can be used to systemically deliver a therapeutic gene into brain tumor. See Materials and Methods #10 for more details.

Example 10. Intracerebral Administration of
AAV.CPP.21

In addition to systemic administration (such as in Example 2), an AAV as described herein was administered locally into the mouse brain. Intracerebral injection of AAV9-H2B-RFP and AAV.CPP.21-H2B-RFP (FIG. 13) resulted in more widespread and higher-intensity RFP signal in AAV.CPP.21-treated brain sections vs. AAV9-treated ones. See Materials and Methods #4 for more details.

Other Embodiments

It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

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<221> NAME/KEY: MOD_RES
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: K, R, H, D, or E
<220> FEATURE:
<223> OTHER INFORMATION: See specification as filed for detailed
description of substitutions and preferred embodiments

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<400> SEQUENCE: 74

Thr Xaa Xaa Xaa Xaa Xaa
1 5

<210> SEQ ID NO 75
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (1)..(4)
<223> OTHER INFORMATION: V, A, L, I, G, P, S, T, or M
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (5)..(5)
<223> OTHER INFORMATION: K, R, H, D, or E
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: E or D
<220> FEATURE:
<223> OTHER INFORMATION: See specification as filed for detailed description of substitutions and preferred embodiments

<400> SEQUENCE: 75

Xaa Xaa Xaa Xaa Xaa Xaa
1 5

<210> SEQ ID NO 76
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (1)..(1)
<223> OTHER INFORMATION: F, L, W, or Y
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(6)
<223> OTHER INFORMATION: V, A, L, I, G, P, S, T, or M
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: K, R, H, D, or E
<220> FEATURE:
<223> OTHER INFORMATION: See specification as filed for detailed description of substitutions and preferred embodiments

<400> SEQUENCE: 76

Xaa Thr Xaa Xaa Xaa Xaa Xaa
1 5

<210> SEQ ID NO 77
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (2)..(5)
<223> OTHER INFORMATION: V, A, L, I, G, P, S, T, or M
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (6)..(6)

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<223> OTHER INFORMATION: K, R, H, D, or E
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: E or D
<220> FEATURE:
<223> OTHER INFORMATION: See specification as filed for detailed
description of substitutions and preferred embodiments

<400> SEQUENCE: 77

Thr Xaa Xaa Xaa Xaa Xaa Xaa
1 5

<210> SEQ ID NO 78
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (1)..(4)
<223> OTHER INFORMATION: V, A, L, I, G, P, S, T, or M
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (5)..(5)
<223> OTHER INFORMATION: K, R, H, D, or E
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: E or D
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: A or I
<220> FEATURE:
<223> OTHER INFORMATION: See specification as filed for detailed
description of substitutions and preferred embodiments

<400> SEQUENCE: 78

Xaa Xaa Xaa Xaa Xaa Xaa Xaa
1 5

<210> SEQ ID NO 79
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (2)..(2)
<223> OTHER INFORMATION: S or P
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: A, M, or T
<220> FEATURE:
<223> OTHER INFORMATION: See specification as filed for detailed
description of substitutions and preferred embodiments

<400> SEQUENCE: 79

Val Xaa Xaa Leu
1

<210> SEQ ID NO 80
<211> LENGTH: 5
<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: S or P
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (4)..(4)
<223> OTHER INFORMATION: A, M, or T
<220> FEATURE:
<223> OTHER INFORMATION: See specification as filed for detailed
      description of substitutions and preferred embodiments

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<400> SEQUENCE: 80

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Thr Val Xaa Xaa Leu
1             5

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<210> SEQ ID NO 81
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: S or P
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (4)..(4)
<223> OTHER INFORMATION: A, M, or T
<220> FEATURE:
<223> OTHER INFORMATION: See specification as filed for detailed
      description of substitutions and preferred embodiments

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<400> SEQUENCE: 81

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Thr Val Xaa Xaa Leu Lys
1             5

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<210> SEQ ID NO 82
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: S or P
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (4)..(4)
<223> OTHER INFORMATION: A, M, or T
<220> FEATURE:
<223> OTHER INFORMATION: See specification as filed for detailed
      description of substitutions and preferred embodiments

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<400> SEQUENCE: 82

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Thr Val Xaa Xaa Leu Phe Lys
1             5

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<210> SEQ ID NO 83
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

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<400> SEQUENCE: 83

Lys Leu Ala Ser Val Thr
1 5

<210> SEQ ID NO 84

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 84

Lys Phe Leu Ala Ser Val Thr
1 5

<210> SEQ ID NO 85

<211> LENGTH: 736

<212> TYPE: PRT

<213> ORGANISM: Adeno-associated virus 9

<400> SEQUENCE: 85

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1 5 10 15

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
20 25 30

Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
35 40 45

Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
50 55 60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65 70 75 80

Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
85 90 95

Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
100 105 110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
115 120 125

Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
130 135 140

Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
145 150 155 160

Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
165 170 175

Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
180 185 190

Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly
195 200 205

Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser
210 215 220

Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile
225 230 235 240

Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu
245 250 255

Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn
260 265 270

Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
275						280				285					
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
290						295				300					
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
305				310						315				320	
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
		325						330						335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
		340				345						350			
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
355						360						365			
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
370						375				380					
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
385				390						395				400	
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
		405						410						415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
		420				425						430			
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
435						440						445			
Lys	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser
450						455				460					
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
465				470						475				480	
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
		485						490						495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
		500						505				510			
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
515						520						525			
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
530						535						540			
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545				550						555				560	
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
		565						570						575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Ala	Gln	Ala	Gln
		580						585				590			
Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	Leu	Pro	Gly	Met	Val	Trp	Gln
595						600						605			
Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	Pro	His
610						615						620			
Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	Gly	Met
625				630						635				640	
Lys	His	Pro													

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Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn
 690 695 700
 Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Asn Thr Glu Gly Val
 705 710 715 720
 Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu
 725 730 735

 <210> SEQ ID NO 86
 <211> LENGTH: 609
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

 <400> SEQUENCE: 86
 Met Ser Gly Trp Glu Ser Tyr Tyr Lys Thr Glu Gly Asp Glu Glu Ala
 1 5 10 15
 Glu Glu Glu Gln Glu Glu Asn Leu Glu Ala Ser Gly Asp Tyr Lys Tyr
 20 25 30
 Ser Gly Arg Asp Ser Leu Ile Phe Leu Val Asp Ala Ser Lys Ala Met
 35 40 45
 Phe Glu Ser Gln Ser Glu Asp Glu Leu Thr Pro Phe Asp Met Ser Ile
 50 55 60
 Gln Cys Ile Gln Ser Val Tyr Ile Ser Lys Ile Ile Ser Ser Asp Arg
 65 70 75 80
 Asp Leu Leu Ala Val Val Phe Tyr Gly Thr Glu Lys Asp Lys Asn Ser
 85 90 95
 Val Asn Phe Lys Asn Ile Tyr Val Leu Gln Glu Leu Asp Asn Pro Gly
 100 105 110
 Ala Lys Arg Ile Leu Glu Leu Asp Gln Phe Lys Gly Gln Gln Gly Gln
 115 120 125
 Lys Arg Phe Gln Asp Met Met Gly His Gly Ser Asp Tyr Ser Leu Ser
 130 135 140
 Glu Val Leu Trp Val Cys Ala Asn Leu Phe Ser Asp Val Gln Phe Lys
 145 150 155 160
 Met Ser His Lys Arg Ile Met Leu Phe Thr Asn Glu Asp Asn Pro His
 165 170 175
 Gly Asn Asp Ser Ala Lys Ala Ser Arg Ala Arg Thr Lys Ala Gly Asp
 180 185 190
 Leu Arg Asp Thr Gly Ile Phe Leu Asp Leu Met His Leu Lys Lys Pro
 195 200 205
 Gly Gly Phe Asp Ile Ser Leu Phe Tyr Arg Asp Ile Ile Ser Ile Ala
 210 215 220
 Glu Asp Glu Asp Leu Arg Val His Phe Glu Glu Ser Ser Lys Leu Glu
 225 230 235 240
 Asp Leu Leu Arg Lys Val Arg Ala Lys Glu Thr Arg Lys Arg Ala Leu
 245 250 255
 Ser Arg Leu Lys Leu Lys Leu Asn Lys Asp Ile Val Ile Ser Val Gly
 260 265 270
 Ile Tyr Asn Leu Val Gln Lys Ala Leu Lys Pro Pro Pro Ile Lys Leu
 275 280 285
 Tyr Arg Glu Thr Asn Glu Pro Val Lys Thr Lys Thr Arg Thr Phe Asn
 290 295 300
 Thr Ser Thr Gly Gly Leu Leu Leu Pro Ser Asp Thr Lys Arg Ser Gln
 305 310 315 320

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Ile Tyr Gly Ser Arg Gln Ile Ile Leu Glu Lys Glu Glu Thr Glu Glu
      325                      330                      335

Leu Lys Arg Phe Asp Asp Pro Gly Leu Met Leu Met Gly Phe Lys Pro
      340                      345                      350

Leu Val Leu Leu Lys Lys His His Tyr Leu Arg Pro Ser Leu Phe Val
      355                      360                      365

Tyr Pro Glu Glu Ser Leu Val Ile Gly Ser Ser Thr Leu Phe Ser Ala
      370                      375                      380

Leu Leu Ile Lys Cys Leu Glu Lys Glu Val Ala Ala Leu Cys Arg Tyr
      385                      390                      395                      400

Thr Pro Arg Arg Asn Ile Pro Pro Tyr Phe Val Ala Leu Val Pro Gln
      405                      410                      415

Glu Glu Glu Leu Asp Asp Gln Lys Ile Gln Val Thr Pro Pro Gly Phe
      420                      425                      430

Gln Leu Val Phe Leu Pro Phe Ala Asp Asp Lys Arg Lys Met Pro Phe
      435                      440                      445

Thr Glu Lys Ile Met Ala Thr Pro Glu Gln Val Gly Lys Met Lys Ala
      450                      455                      460

Ile Val Glu Lys Leu Arg Phe Thr Tyr Arg Ser Asp Ser Phe Glu Asn
      465                      470                      475                      480

Pro Val Leu Gln Gln His Phe Arg Asn Leu Glu Ala Leu Ala Leu Asp
      485                      490                      495

Leu Met Glu Pro Glu Gln Ala Val Asp Leu Thr Leu Pro Lys Val Glu
      500                      505                      510

Ala Met Asn Lys Arg Leu Gly Ser Leu Val Asp Glu Phe Lys Glu Leu
      515                      520                      525

Val Tyr Pro Pro Asp Tyr Asn Pro Glu Gly Lys Val Thr Lys Arg Lys
      530                      535                      540

His Asp Asn Glu Gly Ser Gly Ser Lys Arg Pro Lys Val Glu Tyr Ser
      545                      550                      555                      560

Glu Glu Glu Leu Lys Thr His Ile Ser Lys Gly Thr Leu Gly Lys Phe
      565                      570                      575

Thr Val Pro Met Leu Lys Glu Ala Cys Arg Ala Tyr Gly Leu Lys Ser
      580                      585                      590

Gly Leu Lys Lys Gln Glu Leu Leu Glu Ala Leu Thr Lys His Phe Gln
      595                      600                      605

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Asp

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<210> SEQ ID NO 87
<211> LENGTH: 608
<212> TYPE: PRT
<213> ORGANISM: Mus sp.

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<400> SEQUENCE: 87

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Met Ser Glu Trp Glu Ser Tyr Tyr Lys Thr Glu Gly Glu Glu Glu Glu
 1          5          10          15

Glu Glu Glu Glu Ser Pro Asp Thr Gly Gly Glu Tyr Lys Tyr Ser Gly
 20          25          30

Arg Asp Ser Leu Ile Phe Leu Val Asp Ala Ser Arg Ala Met Phe Glu
 35          40          45

Ser Gln Gly Glu Asp Glu Leu Thr Pro Phe Asp Met Ser Ile Gln Cys
 50          55          60

Ile Gln Ser Val Tyr Thr Ser Lys Ile Ile Ser Ser Asp Arg Asp Leu
 65          70          75          80

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Leu	Ala	Val		Phe	Tyr	Gly	Thr	Glu	Lys	Asp	Lys	Asn	Ser	Val	Asn	
				85					90				95			
Phe	Lys	Asn	Ile	Tyr	Val	Leu	Gln	Asp	Leu	Asp	Asn	Pro	Gly	Ala	Lys	
				100					105				110			
Arg	Val	Leu	Glu	Leu	Asp	Gln	Phe	Lys	Gly	Gln	Gln	Gly	Lys	Lys	His	
				115					120				125			
Phe	Arg	Asp	Thr	Val	Gly	His	Gly	Ser	Asp	Tyr	Ser	Leu	Ser	Glu	Val	
				130									140			
Leu	Trp	Val	Cys	Ala	Asn	Leu	Phe	Ser	Asp	Val	Gln	Leu	Lys	Met	Ser	
				145									155			
His	Lys	Arg	Ile	Met	Leu	Phe	Thr	Asn	Glu	Asp	Asp	Pro	His	Gly	Arg	
				165									175			
Asp	Ser	Ala	Lys	Ala	Ser	Arg	Ala	Arg	Thr	Lys	Ala	Ser	Asp	Leu	Arg	
				180									185			
Asp	Thr	Gly	Ile	Phe	Leu	Asp	Leu	Met	His	Leu	Lys	Lys	Pro	Gly	Gly	
				195									200			
Phe	Asp	Val	Ser	Val	Phe	Tyr	Arg	Asp	Ile	Ile	Thr	Thr	Ala	Glu	Asp	
				210									215			
Glu	Asp	Leu	Gly	Val	His	Phe	Glu	Glu	Ser	Ser	Lys	Leu	Glu	Asp	Leu	
				225									230			
Leu	Arg	Lys	Val	Arg	Ala	Lys	Glu	Thr	Lys	Lys	Arg	Val	Leu	Ser	Arg	
				245									250			
Leu	Lys	Phe	Lys	Leu	Gly	Glu	Asp	Val	Val	Leu	Met	Val	Gly	Ile	Tyr	
				260									265			
Asn	Leu	Val	Gln	Lys	Ala	Asn	Lys	Pro	Phe	Pro	Val	Arg	Leu	Tyr	Arg	
				275									280			
Glu	Thr	Asn	Glu	Pro	Val	Lys	Thr	Lys	Thr	Arg	Thr	Phe	Asn	Val	Asn	
				290									295			
Thr	Gly	Ser	Leu	Leu	Leu	Pro	Ser	Asp	Thr	Lys	Arg	Ser	Leu	Thr	Tyr	
				305									310			
Gly	Thr	Arg	Gln	Ile	Val	Leu	Glu	Lys	Glu	Glu	Thr	Glu	Glu	Leu	Lys	
				325									330			
Arg	Phe	Asp	Glu	Pro	Gly	Leu	Ile	Leu	Met	Gly	Phe	Lys	Pro	Thr	Val	
				340									345			
Met	Leu	Lys	Lys	Gln	His	Tyr	Leu	Arg	Pro	Ser	Leu	Phe	Val	Tyr	Pro	
				355									360			
Glu	Glu	Ser	Leu	Val	Ser	Gly	Ser	Ser	Thr	Leu	Phe	Ser	Ala	Leu	Leu	
				370									375			
Thr	Lys	Cys	Val	Glu	Lys	Glu	Val	Ile	Ala	Val	Cys	Arg	Tyr	Thr	Pro	
				385									390			
Arg	Lys	Asn	Val	Ser	Pro	Tyr	Phe	Val	Ala	Leu	Val	Pro	Gln	Glu	Glu	
				405									410			
Glu	Leu	Asp	Asp	Gln	Asn	Ile	Gln	Val	Thr	Pro	Gly	Gly	Phe	Gln	Leu	
				420									425			
Val	Phe	Leu	Pro	Tyr	Ala	Asp	Asp	Lys	Arg	Lys	Val	Pro	Phe	Thr	Glu	
				435									440			
Lys	Val	Thr	Ala	Asn	Gln	Glu	Gln	Ile	Asp	Lys	Met	Lys	Ala	Ile	Val	
				450									455			
Gln	Lys	Leu	Arg	Phe	Thr	Tyr	Arg	Ser	Asp	Ser	Phe	Glu	Asn	Pro	Val	
				465									470			
Leu	Gln	Gln	His	Phe	Arg	Asn	Leu	Glu	Ala	Leu	Ala	Leu	Asp	Met	Met	
				485									490			

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Glu Ser Glu Gln Val Val Asp Leu Thr Leu Pro Lys Val Glu Ala Ile
 500 505 510
 Lys Lys Arg Leu Gly Ser Leu Ala Asp Glu Phe Lys Glu Leu Val Tyr
 515 520 525
 Pro Pro Gly Tyr Asn Pro Glu Gly Lys Val Ala Lys Arg Lys Gln Asp
 530 535 540
 Asp Glu Gly Ser Thr Ser Lys Lys Pro Lys Val Glu Leu Ser Glu Glu
 545 550 555 560
 Glu Leu Lys Ala His Phe Arg Lys Gly Thr Leu Gly Lys Leu Thr Val
 565 570 575
 Pro Thr Leu Lys Asp Ile Cys Lys Ala His Gly Leu Lys Ser Gly Pro
 580 585 590
 Lys Lys Gln Glu Leu Leu Asp Ala Leu Ile Arg His Leu Glu Lys Asn
 595 600 605

<210> SEQ ID NO 88
 <211> LENGTH: 608
 <212> TYPE: PRT
 <213> ORGANISM: Rattus sp.

<400> SEQUENCE: 88

Met Ser Glu Trp Glu Ser Tyr Tyr Lys Thr Glu Gly Glu Glu Glu
 1 5 10 15
 Glu Glu Glu Gln Ser Pro Asp Thr Asn Gly Glu Tyr Lys Tyr Ser Gly
 20 25 30
 Arg Asp Ser Leu Ile Phe Leu Val Asp Ala Ser Arg Ala Met Phe Glu
 35 40 45
 Ser Gln Gly Glu Asp Glu Leu Thr Pro Phe Asp Met Ser Ile Gln Cys
 50 55 60
 Ile Gln Ser Val Tyr Thr Ser Lys Ile Ile Ser Ser Asp Arg Asp Leu
 65 70 75 80
 Leu Ala Val Val Phe Tyr Gly Thr Glu Lys Asp Lys Asn Ser Val Asn
 85 90 95
 Phe Lys Ser Ile Tyr Val Leu Gln Asp Leu Asp Asn Pro Gly Ala Lys
 100 105 110
 Arg Val Leu Glu Leu Asp Arg Phe Lys Gly Gln Gln Gly Lys Lys His
 115 120 125
 Phe Arg Asp Thr Ile Gly His Gly Ser Asp Tyr Ser Leu Ser Glu Val
 130 135 140
 Leu Trp Val Cys Ala Asn Leu Phe Ser Asp Val Gln Phe Lys Met Ser
 145 150 155 160
 His Lys Arg Ile Met Leu Phe Thr Asn Glu Asp Asp Pro His Gly Asn
 165 170 175
 Asp Ser Ala Lys Ala Ser Arg Ala Arg Thr Lys Ala Ser Asp Leu Arg
 180 185 190
 Asp Thr Gly Ile Phe Leu Asp Leu Met His Leu Lys Lys Arg Gly Gly
 195 200 205
 Phe Asp Val Ser Leu Phe Tyr Arg Asp Ile Ile Ser Ile Ala Glu Asp
 210 215 220
 Glu Asp Leu Gly Val His Phe Glu Glu Ser Ser Lys Leu Glu Asp Leu
 225 230 235 240
 Leu Arg Lys Val Arg Ala Lys Glu Thr Lys Lys Arg Val Leu Ser Arg
 245 250 255
 Leu Lys Phe Lys Leu Gly Lys Asp Val Ala Leu Met Val Gly Val Tyr
 260 265 270

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Asn Leu Val Gln Lys Ala Asn Lys Pro Phe Pro Val Arg Leu Tyr Arg
  275                      280                      285

Glu Thr Asn Glu Pro Val Lys Thr Lys Thr Arg Thr Phe Asn Val Asn
  290                      295                      300

Thr Gly Ser Leu Leu Leu Pro Ser Asp Thr Lys Arg Ser Leu Thr Phe
  305                      310                      315                      320

Gly Thr Arg Gln Ile Val Leu Glu Lys Glu Glu Thr Glu Glu Leu Lys
  325                      330                      335

Arg Phe Asp Glu Pro Gly Leu Ile Leu Met Gly Phe Lys Pro Met Val
  340                      345                      350

Met Leu Lys Asn His His Tyr Leu Arg Pro Ser Leu Phe Leu Tyr Pro
  355                      360                      365

Glu Glu Ser Leu Val Asn Gly Ser Ser Thr Leu Phe Ser Ala Leu Leu
  370                      375                      380

Thr Lys Cys Val Glu Lys Glu Val Ile Ala Val Cys Arg Tyr Thr Ala
  385                      390                      395                      400

Arg Lys Asn Val Ser Pro Tyr Phe Val Ala Leu Val Pro Gln Glu Glu
  405                      410                      415

Glu Leu Asp Asp Gln Asn Ile Gln Val Thr Pro Ala Gly Phe Gln Leu
  420                      425                      430

Val Phe Leu Pro Tyr Ala Asp Asp Lys Arg Lys Val Pro Phe Thr Glu
  435                      440                      445

Lys Val Met Ala Asn Pro Glu Gln Ile Asp Lys Met Lys Ala Ile Val
  450                      455                      460

Gln Lys Leu Arg Phe Thr Tyr Arg Ser Asp Ser Phe Glu Asn Pro Val
  465                      470                      475                      480

Leu Gln Gln His Phe Arg Asn Leu Glu Ala Leu Ala Leu Asp Met Met
  485                      490                      495

Glu Ser Glu Gln Val Val Asp Leu Thr Leu Pro Lys Val Glu Ala Ile
  500                      505                      510

Lys Lys Arg Leu Gly Ser Leu Ala Asp Glu Phe Lys Glu Leu Val Tyr
  515                      520                      525

Pro Pro Gly Tyr Asn Pro Glu Gly Lys Ile Ala Lys Arg Lys Ala Asp
  530                      535                      540

Asn Glu Gly Ser Ala Ser Lys Lys Pro Lys Val Glu Leu Ser Glu Glu
  545                      550                      555                      560

Glu Leu Lys Asp Leu Phe Ala Lys Gly Thr Leu Gly Lys Leu Thr Val
  565                      570                      575

Pro Ala Leu Arg Asp Ile Cys Lys Ala Tyr Gly Leu Lys Ser Gly Pro
  580                      585                      590

Lys Lys Gln Glu Leu Leu Glu Ala Leu Ser Arg His Leu Glu Lys Asn
  595                      600                      605

```

<210> SEQ ID NO 89

<211> LENGTH: 742

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 89

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Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
  1           5           10           15

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-continued

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
 145 150 155 160
 Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
 165 170 175
 Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
 180 185 190
 Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly
 195 200 205
 Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser
 210 215 220
 Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile
 225 230 235 240
 Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu
 245 250 255
 Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn
 260 265 270
 Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg
 275 280 285
 Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn
 290 295 300
 Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn Ile
 305 310 315 320
 Gln Val Lys Glu Val Thr Asp Asn Asn Gly Val Lys Thr Ile Ala Asn
 325 330 335
 Asn Leu Thr Ser Thr Val Gln Val Phe Thr Asp Ser Asp Tyr Gln Leu
 340 345 350
 Pro Tyr Val Leu Gly Ser Ala His Glu Gly Cys Leu Pro Pro Phe Pro
 355 360 365
 Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asp
 370 375 380
 Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe
 385 390 395 400
 Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu
 405 410 415
 Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu
 420 425 430

-continued

Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser
 435 440 445
 Lys Thr Ile Asn Gly Ser Gly Gln Asn Gln Gln Thr Leu Lys Phe Ser
 450 455 460
 Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro
 465 470 475 480
 Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn
 485 490 495
 Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn
 500 505 510
 Gly Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys
 515 520 525
 Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly
 530 535 540
 Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile
 545 550 555 560
 Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser
 565 570 575
 Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Thr Val Ser Ala
 580 585 590
 Leu Lys Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile Leu
 595 600 605
 Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile
 610 615 620
 Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu
 625 630 635 640
 Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile Lys
 645 650 655
 Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys
 660 665 670
 Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu
 675 680 685
 Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu
 690 695 700
 Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala
 705 710 715 720
 Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg
 725 730 735
 Tyr Leu Thr Arg Asn Leu
 740

<210> SEQ ID NO 90
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 90

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45

Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro
50 55 60															
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp
65 70 75 80															
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala
85 90 95															
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly
100 105 110															
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro
115 120 125															
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg
130 135 140															
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly
145 150 155 160															
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr
165 170 175															
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro
180 185 190															
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly
195 200 205															
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser
210 215 220															
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile
225 230 235 240															
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
245 250 255															
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
260 265 270															
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
275 280 285															
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
290 295 300															
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
305 310 315 320															
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
325 330 335															
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
340 345 350															
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
355 360 365															
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
370 375 380															
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
385 390 395 400															
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
405 410 415															
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
420 425 430															
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
435 440 445															
Lys	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser
450 455 460															

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Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
465					470					475					480
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
				485					490					495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500					505					510		
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
		515					520					525			
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
	530					535					540				
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
			565						570					575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Thr	Val	Ser	Ala
			580					585					590		
Leu	Phe	Lys	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile
		595					600					605			
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro
	610					615					620				
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro
625					630					635					640
Leu	Met	Gly	Gly	Phe	Gly	Met	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile
				645					650					655	
Lys	Asn	Thr	Pro	Val	Pro	Ala	Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp
			660					665					670		
Lys	Leu	Asn	Ser	Phe	Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val
		675					680					685			
Glu	Ile	Glu	Trp	Glu	Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro
	690					695					700				
Glu	Ile	Gln	Tyr	Thr	Ser	Asn	Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe
705					710					715					720
Ala	Val	Asn	Thr	Glu	Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr
				725					730					735	
Arg	Tyr	Leu	Thr	Arg	Asn	Leu									
				740											

<210> SEQ ID NO 91
 <211> LENGTH: 21
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 primer

<400> SEQUENCE: 91

ggaaccacctgtgatggagt t

21

<210> SEQ ID NO 92
 <211> LENGTH: 16
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 primer

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<400> SEQUENCE: 92

cggcctcagt gagcga

16

<210> SEQ ID NO 93

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:
SynB1 sequence

<400> SEQUENCE: 93

Arg Gly Gly Arg Leu Ser Tyr Ser Arg Arg Arg Phe Ser Thr Ser Thr

1 5 10 15

Gly Arg

<210> SEQ ID NO 94

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 94

His Ala Arg Ile Lys Pro Thr Phe Arg Arg Leu Lys Trp Lys Tyr Lys

1 5 10 15

Gly Lys Phe Trp

20

<210> SEQ ID NO 95

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 95

Pro Leu Ser Ser Ile Phe Ser Arg Ile Gly Asp Pro

1 5 10

<210> SEQ ID NO 96

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 96

Ala Gly Tyr Leu Leu Gly Lys Ile Asn Leu Lys Ala Leu Ala Ala Leu

1 5 10 15

Ala Lys Lys Ile Leu

20

<210> SEQ ID NO 97

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

-continued

<400> SEQUENCE: 97

Val Arg Leu Pro Pro Pro Val Arg Leu Pro Pro Pro Val Arg Leu Pro
1 5 10 15

Pro Pro

<210> SEQ ID NO 98

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 98

Val Glu Leu Pro Pro Pro Val Glu Leu Pro Pro Pro Val Glu Leu Pro
1 5 10 15

Pro Pro

<210> SEQ ID NO 99

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 99

Lys Gly Thr Tyr Lys Lys Lys Leu Met Arg Ile Pro Leu Lys Gly Thr
1 5 10 15

<210> SEQ ID NO 100

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 100

Pro Pro Arg Pro Pro Arg Pro Pro Arg
1 5

<210> SEQ ID NO 101

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 101

Pro Pro Arg Pro Pro Arg Pro Pro Arg Pro Pro Arg Pro Pro Arg
1 5 10 15

<210> SEQ ID NO 102

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 102

Arg Arg Arg Arg Arg Arg Arg Arg
1 5

-continued

```

<210> SEQ ID NO 103
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

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<400> SEQUENCE: 103

```

```

Leu Arg Arg Glu Arg Gln Ser Arg Leu Arg Arg Glu Arg Gln Ser Arg
1           5              10             15

```

```

<210> SEQ ID NO 104
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Human immunodeficiency virus 1

```

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<400> SEQUENCE: 104

```

```

Arg Lys Lys Arg Arg Gln Arg Arg
1           5

```

What is claimed is:

1. An AAV capsid protein comprising a targeting sequence that comprises TVSALFK (SEQ ID NO:8); TVSALK (SEQ ID NO:4); KLASVT (SEQ ID NO:83); or KFLASVT (SEQ ID NO:84), wherein the targeting sequence is inserted in a position corresponding to amino acids 588 and 589 of SEQ ID NO:85.

2. The AAV capsid protein of claim 1, comprising an amino acid sequence that comprises TVSALK (SEQ ID NO:4).

3. The AAV capsid protein of claim 1, comprising an amino acid sequence that comprises TVSALFK (SEQ ID NO:8).

4. The AAV capsid protein of claim 1, wherein the AAV is AAV9.

25 5. The AAV capsid protein of claim 1, comprising AAV9 VP1.

6. An AAV comprising the capsid protein of claim 1.

7. The AAV of claim 6, further comprising a transgene.

30 8. The AAV capsid protein of claim 1, comprising an amino acid sequence of SEQ ID NO: 89 or SEQ ID NO: 90.

9. The AAV of claim 7, wherein the transgene is a therapeutic transgene.

35 10. The AAV of claim 7, wherein the transgene comprises a non-coding RNA.

11. The AAV of claim 7, wherein the transgene encodes an anti-cancer agent.

* * * * *